When do local repairs of a P_3 packing combine into a cluster editing?

(working title; draft skeleton)

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam
vinh.dq4@buv.edu.vn

Abstract

Li, Pilipczuk and Sorge study Cluster Editing parameterized by the excess ℓ of the solution size over a modification-disjoint packing H of induced P_3's. We ask which interaction structure of H makes this parameterization tractable, and use the treewidth w of the intersection graph $I(H)$ of the packing as the measure. We prove: (i) for $\ell = 0$ the problem is a CSP with domain size 3 whose constraint scopes are cliques of $I(H)$, hence solvable in $3^{w+1}n^{O(1)}$ given a tree decomposition; (ii) for every ℓ it is solvable in $7^{w+1}n^{2\ell+O(1)}$ given a tree decomposition (XP in ℓ, FPT in w), and domain size 7 cannot be lowered in the natural encoding; (iii) under ETH there is no $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$ algorithm even for $\ell = 0$ (exponent only, no base); (iv) w alone does not suffice (NP-hard for $w = 0$ with unbounded ℓ), and $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ are incomparable, with a tradeoff $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$ on stars; (v) adjacent true twins lie in one cluster in every optimum, and if H is maximal among pair-disjoint packings (no induced P_3 has all three pairs unpacked), then in *every* optimum no cluster contains more than one whole twin class of vertices outside $V(H)$, so no edit has both endpoints outside $V(H)$; we derive a normal form (frozen classes, a shrink rule with a tight threshold, a size bound $|R| \leq (\ell+1)|V(H)|$, a classification of the partners of edits touching R) and show by a five-vertex example that pair-maximality cannot be weakened to maximality among vertex-disjoint packings; (vi) CEaMP is solvable in time $1.62^{\binom{s}{2}+1)\ell}n^{O(1)} + n3^s$ and has a kernel with at most $(s^2-s+2)\ell$ vertices, where s is the largest number of vertices spanned by a connected component of $I(H)$; in particular $I(H)$ a perfect matching is FPT in ℓ. This is proved from a rule of van Bevern, Froese and Komusiewicz after verifying its hypotheses at the level of components; the parts that rest on their published proofs are listed explicitly. The main question, whether the problem is fixed-parameter tractable in $\ell + w$, is *open*; we do not prove or disprove it. (vii) For every packing H and every YES instance we prove counting lemmas on the structure of solutions: maximalising H by pair-disjoint induced P_3's costs at most $+|M| \leq \ell$ in treewidth (tight on stars); the hard non-adjacent pairs among the non-packed neighbours of a vertex have matching number at most ℓ, so the independence number of that neighbourhood is at most $2\ell + c + 1$; pieces of a solution that share a cluster are small ($|C| \leq \max\{D + u + 1, 2D, 2u\}$ for a cluster with at least two pieces); and the graph of packed pairs has treewidth at most $3w+2$. A clearly labelled heuristic discussion explains why hub and grid-tiling encodings do not give hardness; it is not a theorem. These lemmas give neither an FPT algorithm in $\ell + w$ nor hardness.

(viii) *Locality of inserted pairs.* In an optimal solution every cluster has diameter at most 2 in G; with pair-maximality every inserted pair ab has a middle vertex z and is either of type T1 (both steps soft, in two distinct members through z) or of type T2 (one step a kept hard edge inside a piece, one step soft). Pieces that share a cluster and are joined by *no* soft pair have all cross pairs of type T1, hence lie in one bag of a decomposition of $I(H)$ and have at most $4c(c-1)$ partners per vertex; merges of pieces that are joined by a soft pair can be of type T2 (examples with $n = 7$ and $n = 8$), so this locality is proved only for soft-free merges. Two hard vertices in one cluster agree on their hard neighbourhoods up to ℓ vertices.

(ix) *The cluster-identity route.* For a pair-maximal H we prove a normal form of optimal solutions (clusters touching an edited hard pair have at most $2D + \ell + 1$ vertices and are connected; all other clusters are unions of whole hard components) and that one endpoint of an edited hard pair determines a set of at most $f(\ell, D)$ candidates for the other. Consequences: an algorithm in time $n^{\ell} f(\ell, w) 7^{w+\ell+1} n^{O(1)}$ (the exponent of (ii) halved) and an FPT algorithm in (ℓ, D, ω), where ω is the treewidth of the graph of edges and packed pairs after removing hard components of large hard degree. However, ω is not bounded by any function of (ℓ, w) (a cubic-expander family with $\ell = w = 0$, using a cited treewidth bound), so this does not give FPT in $\ell + w$.

What remains is stated exactly: the pointers of type T2, the pointers realised by deleted hard edges, and a reduction lemma (conjectured, not proved) that would bound ω; non-locality of merge pointers at unbounded distance in $I(H)$ is open.

1 Introduction

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph G into a disjoint union of cliques. Every induced P_3 must be hit by an edit, so a packing H of induced P_3's that pairwise share no vertex pair (a *modification-disjoint* packing) gives the lower bound $|H|$ on the solution size. Li, Pilipczuk and Sorge [11] parameterize by the *excess* ℓ over this bound (problem CEaMP): is there a solution of size at most $|H| + \ell$? They show NP-hardness already for $\ell = 0$, with every vertex in at most 23 packed P_3's, and an $n^{2\ell+O(1)}$ algorithm when every vertex lies in at most two packed P_3's; whether the latter case is fixed-parameter tractable is open [11]. For vertex-disjoint packings the problem is FPT in ℓ [14].

The question of this paper is which measurable *interaction structure* of H makes the local repair choices (one edit per packed P_3, in three ways) combine into a global solution in $f(\ell, w)\, n^{O(1)}$ time. We take $w = \mathrm{tw}(I(H))$, where the intersection graph $I(H)$ has the packed P_3's as nodes and joins two of them when they share a vertex.

What is proved. For $\ell = 0$ the problem is a constraint satisfaction problem (CSP) with one variable of domain size 3 per packed P_3 and constraint scopes that are cliques of $I(H)$ (Section 3); dynamic programming over a tree decomposition gives $3^{w+1} n^{O(1)}$. For general ℓ, guessing the at most ℓ edits outside all packed P_3's reduces to the same CSP with domain size 7, giving $7^{w+1} n^{2\ell+O(1)}$ (Section 4). Section 5 proves a strict twin lemma and, for packings H that are maximal among pair-disjoint packings, the structure of the vertices R outside $V(H)$ and the statement (C1) that in every optimum no edit has both endpoints in R; it derives frozen classes, a shrink rule whose threshold is tight, the bound $|R| \le (\ell + 1)|V(H)|$, and a classification of the partners of edits touching R. A five-vertex example shows that the hypothesis cannot be weakened to maximality among vertex-disjoint packings, and a four-cycle shows that edits between R and $V(H)$ cannot be excluded. Section 6 proves Theorem K: with s the maximum number of vertices spanned by a component of $I(H)$, CEaMP runs in $1.62^{(\binom{s}{2}+1)\ell} n^{O(1)} + n3^s$ time and has a kernel with at most $(s^2 - s + 2)\ell$ vertices; $I(H)$ a perfect matching is therefore FPT in ℓ. The proof applies the reduction rule of [14] to the components of $I(H)$ after proving its hypotheses in full; it still relies on their published proof of the rule's soundness, on a Cluster Editing solver and on the $2k$-vertex kernel as quoted there. Section 7 proves counting lemmas for solutions with respect to an arbitrary packing (maximalisation with its cost in treewidth, a matching bound for hubs of deletions, small merges of pieces, and treewidth of the soft graph) and ends with a clearly labelled heuristic discussion of why simple hardness gadgets fail. Section 8 proves that in an optimal solution every cluster has diameter at most 2 and classifies every inserted unpacked pair into two types (T1: two soft steps through a

common neighbour; T2: one kept hard step and one soft step). Pieces that share a cluster and have no soft pair between them have all cross pairs of type T1, which are local in a decomposition of $I(H)$ with at most $4c(c-1)$ partners per vertex; examples with $n = 6, 7, 8$ show that merges of pieces joined by a soft pair can be of type T2, so the locality is proved only for soft-free merges. Section 9 proves a normal form of optimal solutions for pair-maximal packings, a bound on the candidates of one endpoint of an edited hard pair given the other, an $n^{\ell} f(\ell, w) 7^{w+\ell+1}$ algorithm, and an FPT algorithm in (ℓ, D, ω) for the treewidth ω of the graph of edges and packed pairs of the non-*big* part; a family with $\ell = w = 0$ shows that ω is unbounded in (ℓ, w). These sections come with a status table and mark the missing reduction lemma as a conjecture. Under ETH the exponent $\Theta(w)$ cannot be improved to $o(w)$ even at $\ell = 0$; w alone is insufficient; $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ are incomparable (Section 10).

What is not proved. Whether CEaMP is FPT in $\ell + \mathrm{tw}(I(H))$ (or W[1]-hard) is open, already for $w = 1$ with tree components of unbounded size. The algorithm of Section 4 is XP in ℓ, Theorem K needs bounded components, and no hardness is known. The normal form of Section 5 applies to a pair-maximal extension H' of the given packing, whose intersection graph may have larger treewidth, so it does not by itself give a preprocessing for the parameter $(\ell, \mathrm{tw}(I(H)))$. What is left in precise form is: the pointers of type T2, the pointers realised by deleted hard edges, and the reduction lemma ML (Conjecture 64), which would make Theorem 59 an FPT algorithm in $\ell + w$; whether merge pointers can be forced at unbounded distance in $I(H)$ is also open (Section 8.6). The algorithms of Sections 4 and 9 remain XP in ℓ. Section 11 states the open problems.

2 Preliminaries

Graphs are simple and undirected, $n = |V(G)|$. A *pair* is a 2-subset of $V(G)$; for a set S of pairs, $G \triangle S$ flips the adjacency of each pair in S. A graph is a *cluster graph* iff it has no induced P_3, i.e. no vertex triple with exactly two edges. A *solution* is a set S of pairs with $G \triangle S$ a cluster graph; $\mathrm{OPT}(G)$ is the minimum size of a solution.

A *packing* $H = \{P_1, \ldots, P_p\}$ is a set of induced P_3's $P_i = (a_i, b_i, c_i)$ ($a_i b_i, b_i c_i \in E$, $a_i c_i \notin E$). Write $\mathrm{pairs}(P_i)$ for the three pairs inside $V(P_i)$. H is *modification-disjoint* if the sets $\mathrm{pairs}(P_i)$ are pairwise disjoint; this is equivalent to: any two members share at most one vertex (two shared vertices would be a shared pair, and any two vertices of a P_3 form one of its pairs). All packings below are modification-disjoint. A pair is *packed* if it lies in $\bigcup_i \mathrm{pairs}(P_i)$ (its *owner* is then unique) and *unpacked* otherwise. For a vertex u, $Q(u)$ is the set of z with uz packed.

Problem CEaMP. Given G, a modification-disjoint packing H and $\ell \geq 0$, decide whether there is a solution S with $|S| \leq |H| + \ell$ (this is the convention of [11]; ℓ is the excess budget).

The *intersection graph* $I(H)$ has node set H and an edge PP' iff $V(P) \cap V(P') \neq \emptyset$. The P_3's through one vertex form a clique of $I(H)$. Hence (Helly property of tree decompositions) they lie in a common bag, and $c := \max_u |\{P \ni u\}| \leq \mathrm{tw}(I(H)) + 1$ and $|Q(u)| \leq 2c$. We write $w = \mathrm{tw}(I(H))$ with the convention $w = 0$ if $I(H)$ has no edge. A tree decomposition of width w is assumed *given* in the algorithmic statements; computing one costs $2^{O(w^3)} n$ exactly (Bodlaender [2]) or $2^{O(w)} n$ for a factor-2 approximation (Korhonen [10]) (the stated running times are as commonly quoted; we did not re-read them), which would change the bases below (e.g. 7^{w+1} to 7^{2w+2}).

Lemma 1 (Solutions decompose)**.** *Let S be any set of pairs such that $G \triangle S$ is a cluster graph, $x_i = S \cap \mathrm{pairs}(P_i)$ and $U = S \setminus \bigcup_i \mathrm{pairs}(P_i)$. Then every $x_i \neq \emptyset$, $|S| = \sum_i |x_i| + |U|$, and $|U| \leq |S| - |H|$.*

Proof. If $x_i = \emptyset$ the triple $V(P_i)$ keeps its adjacencies in $G \triangle S$ and is an induced P_3. The pair sets are disjoint, so $|S| = \sum |x_i| + |U| \geq |H| + |U|$. □

Remark 2 (Maximality). H is *pair-maximal* if no induced P_3 of G has all three pairs unpacked, i.e. if H is maximal among modification-disjoint packings. This is strictly stronger than maximality among vertex-disjoint packings (Proposition 15), and it is the only notion used in Section 5. If $H' \supseteq H$ adds t further pair-disjoint induced P_3's, then $|H'| + (\ell - t) = |H| + \ell$, so (G, H, ℓ) and $(G, H', \ell - t)$ ask the same question (a NO instance if $\ell < t$, since OPT $\geq |H'|$). A pair-maximal H' is found greedily in polynomial time. Caution: $\mathrm{tw}(I(H'))$ may exceed $\mathrm{tw}(I(H))$.

3 The case $\ell = 0$ is a CSP on the intersection graph

Observation 3. *Let S be a solution with $|S| \leq |H|$. Then S has exactly one pair in each* pairs(P_i) *and no other pair, and $G \triangle S$ is a cluster graph iff no vertex triple is an induced P_3 of $G \triangle S$. Consequently $(G, H, 0)$ is a YES instance iff the following CSP is satisfiable: variables $x_i \in \{1, 2, 3\}$ (which pair of P_i is flipped), and for every vertex triple t the constraint "t is not an induced P_3 after flipping", whose truth depends on the owners' variables of the packed pairs of t only (unpacked pairs keep their status in G).*

Proof. By Lemma 1, $|x_i| \geq 1$ for all i and $|S| = \sum |x_i| + |U| \leq |H|$ force $|x_i| = 1$ and $U = \emptyset$. The rest is the definition of a cluster graph. □

The CSP has $\Theta(n^3)$ constraints, not $O(|H|)$.

Observation 4. *The set of owners of the packed pairs of any triple is a clique of $I(H)$ (at most three nodes). Hence the primal graph of the CSP is a subgraph of $I(H)$ (not every edge of $I(H)$ is witnessed by a triple), and a tree decomposition of $I(H)$ is one of the primal graph.*

Proof. Two pairs of one triple share a vertex v; their owners both contain v, so they coincide or are adjacent in $I(H)$. □

Theorem 5. *Given (G, H) and a tree decomposition of $I(H)$ of width w, CEaMP with $\ell = 0$ is solvable in $3^{w+1} n^{O(1)}$ time.*

Theorem 5 is the case $\ell = 0$ of Theorem 6 below (with a larger base there, 7, because for $\ell > 0$ a packed P_3 may receive several edits); its proof is the same dynamic program with domain $\{1, 2, 3\}$. Observation 4 is what makes the interaction of disjoint P_3's at $\ell = 0$ completely visible in $I(H)$: no transitivity-mediated interaction exists beyond shared vertices.

4 Excess edits: an XP-in-ℓ, FPT-in-w algorithm

Theorem 6. *Given (G, H, ℓ) and a tree decomposition of $I(H)$ of width w, CEaMP is solvable in $7^{w+1} n^{2\ell+O(1)}$ time.*

Proof. Let $\mathcal{D}_i$ be the 7 non-empty subsets of pairs(P_i). By Lemma 1, a solution of size $\leq |H| + \ell$ has the form $S = U \cup \bigcup x_i$ with $x_i \in \mathcal{D}_i$ and U a set of at most ℓ unpacked pairs; there are at most $\sum_{j \leq \ell} \binom{n^2/2}{j} \leq n^{2\ell}$ candidates for U.

Fix U and put $G_U = G \triangle U$. Since U avoids packed pairs, every P_i is still an induced P_3 of G_U with the same pairs, so H is still a modification-disjoint packing. For a triple t, the status of each unpacked pair of t in $G_U \triangle x$ is fixed, and each packed pair is determined by G_U and the variable

of its owner. Hence t gives a constraint C_t ("t is not an induced P_3 in $G_U \triangle x$") whose scope is the owner set of t, a clique of $I(H)$ by Observation 4 (the proof there does not use G itself). Triples with empty scope are tested directly, in $O(n^3)$ time. Seek $x \in \prod \mathcal{D}_i$ satisfying all C_t and minimizing the unary cost $\sum_i |x_i|$.

Let $(T, \{B_s\})$ be the tree decomposition. By the Helly property each clique lies in a bag; assign each constraint to one such bag. For a node s and an assignment α of B_s, let $f_s(\alpha)$ be the minimum cost of $\sum |x_i|$ over variables in the subtree of s, among assignments extending α and satisfying all constraints assigned in the subtree. The standard recurrences (introduce, forget, join; see [4]) compute all f_s in time $7^{w+1}n^{O(1)}$ since there are $O(n^3)$ constraints and each table has $\leq 7^{w+1}$ entries. Accept iff for some U the minimum m_U satisfies $m_U + |U| \leq |H| + \ell$. Soundness: any feasible (U, x) gives the solution $U \cup \bigcup x_i$. Completeness: Lemma 1. □

Remark 7 (Why domain size 7). The empty value is never feasible, so 7 (not 8) suffices. It cannot be lowered to 3: an edited packed P_3 can receive two or three edits in every optimum. For two: $n = 5$, $E = \{02, 04, 12, 23, 24\}$, $H = \{(1, 2, 3)\}$, OPT $= 2$, unique optimum $\{12, 23\}$. For three: $n = 6$, $E = \{01, 02, 03, 04, 12, 14, 15, 24, 25, 35, 45\}$, $H = \{(0, 3, 5)\}$, OPT $= 3$, unique optimum $\{03, 05, 35\}$ (both checked exhaustively over all set partitions). In the notes' random data (computer evidence, $n \leq 9$) about 8–9% of instances force ≥ 2 edits on some packed P_3 and $\approx 0.2\%$ force three. For $c = 2$, $I(H)$ is subcubic but may have unbounded treewidth, so Theorem 6 is a different tradeoff from, not a recovery of, the $n^{2\ell}$ algorithm of [11].

Proposition 8 (Completing a fixed packed assignment)**.** *Let x assign a subset $x_i \subseteq \text{pairs}(P_i)$ to each P_i and let $B = |H| + \ell - \sum |x_i|$ be the excess budget left. Whether there is a set U of at most B unpacked pairs with $G \triangle (x \cup U)$ a cluster graph is decidable in $3^B n^{O(1)}$ time.*

Proof. Search tree on $G_x = G \triangle x$: if the current graph has an induced P_3, some pair of it must be flipped; packed pairs are frozen by x, so branch on its at most three unpacked pairs, reducing B by one (if all its pairs are packed, reject). Depth is at most B; flipping a pair twice is useless, so the search is exhaustive. □

Remark 9. Proposition 8 is a reformulation, not progress towards FPT in (ℓ, w): finding x is the whole problem, and the property "x is completable" is not local in x.

5 Twins, vertices outside the packing, and a normal form

By Lemma 1 the whole difficulty of $\ell > 0$ is the set U of at most ℓ unpacked edits. This section proves what can be said about U without any further assumption on the instance beyond maximality of H (in the pair sense of Remark 2).

Conventions. A *partition* Π of V into *clusters* has *cost* equal to the number of its *edits*: edges between different clusters (deletions) plus non-edges inside a cluster (insertions). Solutions S and partitions correspond ($G \triangle S$ is a cluster graph whose cliques are the clusters), so OPT(G) is the minimum cost and an *optimum* is a partition of minimum cost. $\Pi(v)$ is the cluster of v. Unless said otherwise, H is *pair-maximal* (Remark 2: no induced P_3 of G has all three pairs unpacked) and $R = V \setminus V(H)$; every pair with an end in R is unpacked.

5.1 Twins

Lemma 10 (Twins, strict form)**.** *Let $u \neq v$ be adjacent with $N[u] = N[v]$. For every partition Π with $\Pi(u) \neq \Pi(v)$, one of the two partitions obtained by moving u into $\Pi(v)$, respectively v into $\Pi(u)$, has cost smaller than Π by at least 1; more precisely the two cost changes sum to exactly -2. Hence u, v lie in one cluster in* every *optimum, with no hypothesis on H.*

Proof. For a vertex y and a set $D \not\ni y$ let $f(y, D) = |D \setminus N(y)| - |D \cap N(y)|$. Put $D_u = \Pi(u) \setminus \{u\}$ and $D_v = \Pi(v) \setminus \{v\}$ (possibly empty); $D_u \not\ni v$, $D_v \not\ni u$, and neither contains u or v. Moving u out of $\Pi(u)$ changes the status of the pairs uz, $z \in D_u$: before, uz is an edit iff it is a non-edge; after, iff it is an edge; the change is $-f(u, D_u)$. Moving u into $\Pi(v) = D_v \cup \{v\}$ changes, for $z \in D_v \cup \{v\}$, the status from "edit iff edge" to "edit iff non-edge"; the change is $f(u, D_v \cup \{v\}) = f(u, D_v) - 1$ because $uv \in E$. All other pairs keep their status. So

$$\Delta_1 = -f(u, D_u) + f(u, D_v) - 1, \qquad \Delta_2 = -f(v, D_v) + f(v, D_u) - 1.$$

Since $N(u) \setminus \{v\} = N(v) \setminus \{u\}$ we have $f(u, D) = f(v, D)$ for all $D \subseteq V \setminus \{u, v\}$, hence $\Delta_1 + \Delta_2 = -2$. Thus $\min(\Delta_1, \Delta_2) \leq -1$. □

This is the critical-clique property used in kernelizations of Cluster Editing (Guo [7]; also stated in [14]); the argument above is self-contained and shows that it is strict.

5.2 Viability of an unpacked edit

Lemma 11 (Viability of an unpacked edit)**.** *Let S be any solution, U its unpacked part and $uv \in U$. Let $D(uv)$ be the set of $z \notin Q(u) \cup Q(v) \cup \{u, v\}$ such that $\{u, v, z\}$ is an induced P_3 of $G \triangle \{uv\}$. Then for every $z \in D(uv)$ one of uz, vz lies in U, and $|D(uv)| \leq |U| - 1$.*

Proof. All three pairs of $\{u, v, z\}$ are unpacked ($z \notin Q(u) \cup Q(v)$). If neither uz nor vz is in S, the triple has in $G \triangle S$ the same pattern as in $G \triangle \{uv\}$, an induced P_3. Distinct z give distinct pairs, all different from uv. □

This gives a kernel-style condition (deleting uv needs fewer than ℓ common neighbours outside $Q(u) \cup Q(v)$) but no bound on the number of candidate pairs: all pendant edges are viable.

5.3 The vertices outside $V(H)$

Lemma 12 (Structure of R)**.** *Let H be pair-maximal. For $r, s \in R$: if $rs \in E$ then $N[r] = N[s]$; if $rs \notin E$ then $N(r) \cap N(s) = \emptyset$. Consequently:*

1. *every component X of $G[R]$ (a* class*) is a clique all of whose vertices have the same closed neighbourhood $N[X]$ (they are true twins);*
2. *$A_X := N(X) \setminus X$ is contained in $V(H)$;*
3. *the sets A_X, over the classes X, are pairwise disjoint.*

Proof. Every pair with an end in R is unpacked. Let $rs \in E$ and $z \neq r, s$ be adjacent to exactly one of r, s. Then $\{r, s, z\}$ has exactly two edges, i.e. is an induced P_3 with all three pairs unpacked, contradicting pair-maximality; so $N[r] = N[s]$. If $rs \notin E$ and z is a common neighbour, then $r\,z\,s$ is an induced P_3 whose pairs rz, zs, rs all have an end in R, again a contradiction.

(1) Adjacent vertices of R have equal closed neighbourhoods; along a connected component all closed neighbourhoods coincide, and then any two vertices r, t of the component satisfy $r \in N[r] =$

$N[t]$, so they are adjacent. (2) A neighbour of $x \in X$ lying in R is in the component of x, hence in X. (3) If $a \in A_X \cap A_Y$ for classes $X \neq Y$, then a is a common neighbour of some $x \in X$ and $y \in Y$, which are non-adjacent (different components of $G[R]$); this contradicts the second clause. □

For $x \in X$ and $a \in V(H)$ we thus have $xa \in E$ iff $a \in A_X$; in particular vertices of one class are twins towards $V(H)$ as well. (Lemma 12 fails for vertex-maximal packings, see Proposition 15.)

Lemma 13 (Exchange formula)**.** *Let Π be any partition, X a class with $X \subseteq C$ for a cluster C, $x = |X|$ and $a = |A_X \cap C|$. Let Π' be obtained by moving X out of C into a cluster of its own. Then* $\mathrm{cost}(\Pi') - \mathrm{cost}(\Pi) = x\,(x + 2a - |C|)$.

Proof. Only the pairs between X and $C \setminus X$ change status. By Lemma 12, a vertex of X has exactly a neighbours in $C \setminus X$, namely $A_X \cap C$. Before the move such a pair is an edit iff it is a non-edge: $x(|C| - x - a)$ edits. After the move it is an edit iff it is an edge: xa edits. The difference is $xa - x(|C| - x - a) = x(x + 2a - |C|)$. □

Theorem 14 (C1: no edit inside R)**.** *Let H be pair-maximal. In* every *optimum Π, every cluster meets R in the empty set or in exactly one whole class. Hence no edit of Π has both endpoints in R, and for every class X the cluster $\Pi(X)$ equals $X \cup B_X$ with $B_X := \Pi(X) \cap V(H)$.*

Proof. By Lemma 12(1) and Lemma 10, every class lies inside one cluster (a class of size 1 trivially). Suppose a cluster C contains two distinct classes X_1, X_2. Put $x_i = |X_i| \geq 1$ and $a_i = |A_{X_i} \cap C|$. The sets $X_1, X_2 \subseteq R$ and $A_{X_1} \cap C, A_{X_2} \cap C \subseteq V(H)$ are pairwise disjoint (Lemma 12(2),(3)), so $|C| \geq x_1 + x_2 + a_1 + a_2$. Let Δ_i be the cost change of moving X_i out of C. Since Π is an optimum, $\Delta_1 \geq 0$ and $\Delta_2 \geq 0$, i.e. by Lemma 13 $|C| \leq x_i + 2a_i$ for $i = 1, 2$. With the lower bound on $|C|$: $x_2 + a_2 \leq a_1$ and $x_1 + a_1 \leq a_2$. Adding gives $x_1 + x_2 \leq 0$, absurd. So no cluster contains two classes.

An edit with both ends in R joins two vertices r, s. If they lie in one class, they are adjacent and (by the above) in one cluster, so rs is no edit. If they lie in different classes they are non-adjacent, and rs is an edit iff they share a cluster, which was excluded. □

The strictness of the inequality is what makes the statement hold for every optimum (no ties, and no iteration or termination argument).

Proposition 15 (Pair-maximality cannot be replaced by vertex-maximality)**.** *Let G have vertices $0, \ldots, 4$ and edges $03, 13, 14, 24, 34$, and $H = \{(0, 3, 1)\}$ (pairs $03, 13, 01$). Then H is maximal among vertex-disjoint packings (only the two vertices $2, 4$ remain), $R = \{2, 4\}$, $\mathrm{OPT} = 2$ and the unique optimum is $\{03, 24\}$; it contains the edit 24 with both endpoints in R. So Theorem 14 fails. Here $\{3, 4, 2\}$ is an induced P_3 all of whose pairs are unpacked, so H is not pair-maximal and Lemma 12 fails.*

Proof. The partition $\{\{0\}, \{1, 3, 4\}, \{2\}\}$ has the two edits $03, 24$. Conversely, $0\,3\,1$ and $3\,4\,2$ are induced P_3's with disjoint pair sets $\{03, 13, 01\}$ and $\{34, 24, 23\}$, so $\mathrm{OPT} \geq 2$. That this is the only optimum was checked by enumerating all 52 partitions of the five vertices (computer; not re-derived by hand). □

5.4 Consequences

Let (G, H, ℓ) be a *YES instance* with H pair-maximal, Π any optimum, S its edit set, so $|S| = \mathrm{OPT} \leq |H| + \ell$. Let U be the unpacked edits. By Lemma 1, $|S| - |H| = \sum_i (|x_i| - 1) + |U|$, so $|U| \leq \ell$. Write $q = \max_u |Q(u)| \leq 2c$.

Corollary 16 (Normal form)**.** *For every class X let $B_X = \Pi(X) \cap V(H)$.*

(N1) Types of unpacked edits. *Every unpacked edit is either an* HH-edit *(both ends in $V(H)$) or an* RH-edit $\{x, a\}$, *$x \in X$ a class, $a \in V(H)$; the latter is a deletion if $a \in A_X \setminus B_X$ and an insertion if $a \in B_X \setminus A_X$, and $\{x, a\}$ is edited iff $\{x', a\}$ is, for all $x, x' \in X$. With h the number of HH-edits, $|U| = h + \sum_X |X| \cdot |A_X \triangle B_X| \leq \ell - \sum_i (|x_i| - 1) \leq \ell$.*

(N2) Frozen classes. *If $|X| > \ell$, then $\Pi(X) = X \cup A_X$.*

(N3) Attachment. *Every $a \in V(H)$ lies in A_X for at most one class X. If $z, z' \in A_X$ are non-adjacent then zz' is packed. If $a \in V(H) \setminus A_X$, $z \in A_X$ and $az \in E$ then az is packed; hence $|N(a) \cap A_X| \leq |Q(a)| \leq q$.*

(N4) Shrink rule. *If $|X| \geq \ell + 2$ and $x \in X$, then (G, H, ℓ) is a YES instance iff $(G - x, H, \ell)$ is. This holds for H pair-maximal without the YES hypothesis. If $A_X = \emptyset$ then X is a component of G and may be deleted without changing* OPT.

(N5) Size. *After exhausting the two rules in (N4), every class has at most $\ell + 1$ vertices and at most $|V(H)|$ classes have $A_X \neq \emptyset$, so $|R| \leq (\ell + 1)|V(H)|$ and $n \leq (\ell + 2)|V(H)| \leq 3(\ell + 2)|H|$.*

Proof. (N1) Pairs with an end in R are unpacked; by Theorem 14 none has both ends in R. The vertices of X are twins towards $V(H)$ and lie in one cluster, which gives the all-or-none statement; xa is an edit iff exactly one of $a \in B_X$, $a \in A_X$ holds. Counting gives the identity; the bound is the first paragraph of this subsection.

(N2) If $\Pi(X) \neq X \cup A_X$ then, since $\Pi(X) = X \cup B_X$ (Theorem 14), $A_X \triangle B_X \neq \emptyset$ and (N1) gives $|U| \geq |X| > \ell$, contradicting $|U| \leq \ell$.

(N3) The first claim is Lemma 12(3). Let $x \in X$. If $z, z' \in A_X$ are non-adjacent, $z\,x\,z'$ is an induced P_3 with pairs zx, xz' unpacked, so zz' must be packed (pair-maximality). If $a \in V(H) \setminus A_X$ and $z \in A_X$ with $az \in E$, then $xa \notin E$ and $x\,z\,a$ is an induced P_3 with xz, xa unpacked, so az is packed, i.e. $z \in Q(a)$.

(N4) $x \in R$, so H is still a packing of $G - x$, still pair-maximal (an induced P_3 of $G - x$ is one of G), and $X \setminus \{x\}$ is a class of $G - x$ with the same A_X. ($\Rightarrow$) Restricting a partition of G to $V \setminus x$ does not increase its cost, so $\mathrm{OPT}(G - x) \leq \mathrm{OPT}(G) \leq |H| + \ell$. ($\Leftarrow$) Let Π be an optimum of $G - x$ with cost $\leq |H| + \ell$. Now $|X \setminus x| \geq \ell + 1 > \ell$, so (N2) in $G - x$ gives $\Pi(X \setminus x) = (X \setminus x) \cup A_X$. Add x to this cluster: $N[x] = X \cup A_X$, so every pair at x is satisfied, and the cost does not change. Hence $\mathrm{OPT}(G) \leq |H| + \ell$. If $A_X = \emptyset$, X is a clique component of G, which costs nothing, and H is disjoint from it.

(N5) After the rules each class has at most $\ell + 1$ vertices; the non-empty sets $A_X \subseteq V(H)$ are pairwise disjoint (Lemma 12), and $|V(H)| \leq 3|H|$. Classes with $A_X = \emptyset$ were deleted. □

Remark 17 (Tightness of the threshold in (N4)). For $\ell = 0$ the threshold $|X| \geq 2$ cannot be lowered to $|X| \geq 1$: take the 4-cycle of Proposition 20 below, $H = \{(0, 1, 3)\}$, $X = \{2\}$. Then $(G, H, 0)$ is a NO instance, but $G - 2$ is the induced P_3 0 1 3 itself, so $(G - 2, H, 0)$ is a YES instance. (For larger ℓ the notes report computer evidence of the same tightness; no proof is given here.)

Remark 18 (The rules and the user-supplied packing). The input packing H of CEaMP need not be pair-maximal. Extend it greedily (polynomial time) to a pair-maximal $H' \supseteq H$ with $t = |H'| - |H|$ new members, set $\ell' = \ell - t$ (answer NO if $\ell' < 0$, Remark 2) and apply this section to (G, H', ℓ'). Then (N4), (N5) hold with ℓ' and H'; in particular $n \leq 3(\ell + 2)(|H| + \ell)$ after reduction, since $|H'| \leq |H| + \ell$. But $\mathrm{tw}(I(H'))$ may exceed $\mathrm{tw}(I(H))$, so these rules are a preprocessing for the parameters $(\ell, |H'|)$ or $(\ell, \mathrm{tw}(I(H')))$, not for $(\ell, \mathrm{tw}(I(H)))$ with the given H; and $|V(H')|$ is not bounded by ℓ, so (N5) is not a kernel.

Proposition 19 (Partners of edits touching R)**.** *In the setting of Corollary 16, let X be a class, $B = B_X$. Then:*

(P1) Deletion. *If $\{x, a\}$ is an RH-deletion, $a \in A_X$, then X is determined by a (the unique class with $a \in A_X$): it is a unary flag at a of weight $|X|$. Moreover a is then cut off from every $z \in A_X \cap B$ with $z \notin Q(a)$, each such edge az being an unpacked HH-deletion; so at least $|A_X \cap B| - q$ HH-edits accompany it.*

(P2) No detachment. *If $B \cap A_X = \emptyset$ then $B = \emptyset$.*

(P3) Insertion. *If $a \in B \setminus A_X$ then $B \cap A_X \neq \emptyset$, and for every $z \in B \cap A_X$ either az is packed ($a \in Q(z)$, at most q choices of a per z) or az is an unpacked non-edge, hence an inserted HH-edit.*

(P4) Wide classes. *If $|A_X| \geq \ell + q$ then $B \subseteq A_X$.*

Consequently, every candidate partner $a \in V(H)$ of a class in an RH-edit is unary (P1), local to some z (P3, packed case) or tied to an HH-edit (P3, second case; P1). This classifies the partners; it does not assert that the HH-edits are independent of the RH-edits.

Proof. (P1) Uniqueness is (N3). Let $a \in A_X \setminus B$ and $z \in A_X \cap B$, $z \notin Q(a)$. If $az \notin E$ then az would be packed by (N3), contradicting $z \notin Q(a)$; so $az \in E$, unpacked, and $a \notin \Pi(X) \ni z$, so az is deleted. At most q vertices z lie in $Q(a)$.

(P2) Suppose $B \neq \emptyset = B \cap A_X$. The cluster is $C = X \cup B$ (Theorem 14), $a_X = |A_X \cap C| = 0$, and by Lemma 13 moving X out changes the cost by $|X|(|X| - |X| - |B|) = -|X||B| < 0$, contradicting optimality.

(P3) $B \neq \emptyset$, so $B \cap A_X \neq \emptyset$ by (P2). Let $z \in B \cap A_X$. If $az \in E$ then az is packed by (N3) ($a \in V(H) \setminus A_X$), i.e. $a \in Q(z)$. Otherwise either az is packed (again $a \in Q(z)$) or it is an unpacked non-edge inside the cluster $\Pi(X)$, an insertion.

(P4) Suppose $B \not\subseteq A_X$. Put $\alpha = |A_X \cap B|$, $\delta = |A_X \setminus B|$, $\beta = |B \setminus A_X| \geq 1$. By (N1) there are $|X|\beta$ RH-insertions and $|X|\delta$ RH-deletions. For $b \in B \setminus A_X$ and $z \in A_X \cap B$ with $z \notin Q(b)$, the pair bz is unpacked, lies in a cluster, and is a non-edge by (N3); so it is an unpacked insertion, an HH-edit. There are at least $\beta \max(0, \alpha - q)$ of them. Hence $\ell \geq |U| \geq \beta(|X| + \max(0, \alpha - q)) + |X|\delta$. If $\alpha \geq q$ this gives $\ell \geq 1 + (\alpha - q) + \delta$, i.e. $|A_X| = \alpha + \delta \leq \ell + q - 1$. If $\alpha < q$ it gives $\delta \leq \ell - 1$ and $|A_X| \leq q - 1 + \ell - 1$. Either way $|A_X| < \ell + q$. □

5.5 Where locality stops

Proposition 20 (Strong locality fails)**.** *Let $G = C_4$ with edges $01, 02, 13, 23$ and $H = \{(0, 1, 3)\}$ (pairs $01, 13, 03$; vertex $2 \notin V(H)$). Then H is pair-maximal,* $\mathrm{OPT} = 2 = |H| + 1$*, so $(G, H, 0)$ is a NO instance, and there are exactly three optima $\{03, 12\}$, $\{01, 23\}$, $\{02, 13\}$, each containing an unpacked pair at vertex 2.*

This (checked exhaustively over all partitions, and verified by hand: every induced P_3 of C_4 contains one of the packed pairs $01, 13, 03$) refutes the claim that all unpacked edits of some optimum have both endpoints in $V(H)$; Theorem 14 is what remains true: no edit has both endpoints in R. In the notes' random sample ($n \leq 9$) about 4% of maximal-packing instances with $|R| \geq 2$ force an edit touching R (computer evidence, not a theorem).

Remark 21 (Vertex-disjoint packings and the case $c = 1$). For a vertex-disjoint H ($w = 0$, $c = 1$) FPT in ℓ is the theorem of [14]. One must not derive it from Theorem 14 by treating H as

pair-maximal: a vertex-disjoint H that is maximal among vertex-disjoint packings need not be pair-maximal (Proposition 15; also the path 0 1 2 3 4 with $H = \{(0,1,2)\}$, which is extended by $(2,3,4)$ sharing the vertex 2). The pair-maximal extension H' of Remark 18 is in general no longer vertex-disjoint, so $I(H')$ has edges and the case $w = 0$ is not available for it. We therefore claim nothing about $c = 1$ beyond [14] and Theorem 22.

6 Bounded components of $I(H)$: Theorem K

Let $C_1, \ldots, C_r$ be the connected components of $I(H)$, $p_j = |C_j|$ the number of P_3's in C_j, and $W_j = \bigcup_{P \in C_j} V(P)$. Put $s = \max_j |W_j|$, the largest number of *vertices* spanned by a component. If $s_P = \max_j p_j$ counts P_3's instead, then $s \le 2s_P + 1$: order the P_3's of a component so that each shares a vertex with an earlier one; the first spans 3 vertices and each later one adds at most 2.

Theorem 22 (Theorem K). *CEaMP can be solved in time $1.62^{\binom{s}{2}+1)\ell}\, n^{O(1)} + n\, 3^s$ and admits a kernel with at most $(s^2 - s + 2)\,\ell$ vertices, where s is the maximum number of vertices spanned by a connected component of $I(H)$. (No hypothesis of maximality is used. The kernel is an instance of CEaMP with empty packing.)*

Corollary 23. *If $I(H)$ is a perfect matching (so $w = 1$, $c = 2$), CEaMP is FPT in ℓ: here $s = 5$, the time is $1.62^{11\ell} n^{O(1)}$ and the kernel has at most 22ℓ vertices. More generally components with at most s_P P_3's give Theorem 22 with $s = 2s_P + 1$.*

Proof of Corollary. Two adjacent nodes of $I(H)$ share exactly one vertex (H is modification-disjoint), so $|W_j| = 3 + 3 - 1 = 5$; $s^2 - s + 2 = 22$ and $\binom{5}{2} + 1 = 11$. □

What is cited and what is proved. We use the problem of van Bevern, Froese and Komusiewicz [14]: an instance $(G, \mathcal{G}, k)$ consists of a graph, a set $\mathcal{G}$ of *vertex-disjoint induced subgraphs* G_j with $1 \le \tau(G_j) \le t$, where τ is the cluster-editing optimum, and a budget k; its parameters are t and $\ell_B = k - \sum_j \tau(G_j)$ (their Def. 1.2, Problem 1.3, read in the arXiv text; connectivity of the G_j is not required, and their size enters only through t). We prove below, in full, that the components of $I(H)$ define such an instance and that $\ell_B \le \ell$ (Lemma 24), that the reduction rule can be applied at component level and does not disturb the other components (Lemma 25), and the bound $r \le 2\ell_B$ on the number of irreducible components (Lemma 26; this re-derives, for components, the counting argument of BFK Lemma 3.2 and Lemma 6.5 instead of citing it). *The following rest on the published proofs of [14] and are not reproved here:* (i) soundness of their Reduction Rule 6.2 for an arbitrary induced subgraph $G[W]$ (their Lemma 6.3, whose proof uses only G, W and the optimal solution S of $G[W]$); (ii) the time to test and apply the rule (their Lemma 6.4); (iii) the algorithm of Böcker running in $O(1.62^k + n + m)$ and the kernel with at most $2k$ vertices for Cluster Editing, both as quoted in [14] (we have not opened the original sources).

Lemma 24 (The components form a packing in the sense of [14]). *Let $G_j = G[W_j]$, $\tau_j = \mathrm{OPT}(G_j)$, $k = |H| + \ell$ and $\ell' = k - \sum_j \tau_j$.*

1. *The sets W_j are pairwise disjoint and each G_j is an induced subgraph.*
2. *$p_j \le \tau_j \le \lfloor \binom{s}{2}/2 \rfloor =: t_s$ and $\tau_j \ge 1$.*
3. *$\ell' = \ell - \sum_j (\tau_j - p_j) \le \ell$.*

4. *For every solution S of G and every j, $S_j := S \cap \binom{W_j}{2}$ is a solution of G_j, so $|S_j| \geq \tau_j$; the sets $\binom{W_j}{2}$ are pairwise disjoint, so $|S| \geq \sum_j \tau_j + |S_{\text{ext}}|$ where S_{ext} are the pairs of S inside no W_j. In particular $\text{OPT}(G) \geq \sum_j \tau_j$, and if $\ell' < 0$ then (G, H, ℓ) is a NO instance.*

Proof. (1) A vertex in $W_i \cap W_j$, $i \neq j$, lies in a P_3 of C_i and in a P_3 of C_j; these are adjacent in $I(H)$, contradicting that C_i, C_j are different components. Induced by definition.

(2) The P_3's of C_j are induced P_3's of G_j with pairwise disjoint pair sets inside $\binom{W_j}{2}$. A solution of G_j must edit a pair of each of them (otherwise the triple survives as an induced P_3, as in Lemma 1); hence $\tau_j \geq p_j \geq 1$. For the upper bound, let m_j be the number of edges of G_j; deleting all edges, or inserting all non-edges, yields a cluster graph, so $\tau_j \leq \min(m_j, \binom{|W_j|}{2} - m_j) \leq \binom{|W_j|}{2}/2 \leq \binom{s}{2}/2$.

(3) $|H| = \sum_j p_j$ because the components partition H.

(4) $(G \triangle S)[W_j] = G_j \triangle S_j$ is an induced subgraph of a cluster graph, hence a cluster graph. The rest is counting with disjoint pair sets. □

Thus (G, H, ℓ) is a YES instance iff $(G, \{G_j\}, k)$ is a YES instance of the problem of [14] with $t \leq t_s$ and excess $\ell' \leq \ell$ (answer NO if $\ell' < 0$). Note why components, not single P_3's: a pair uv with $u \in W_j$ and $v \notin W_j$ lies in no W_i at all when v is in R or in another component, hence is not counted in $\sum_j \tau_j$ and is an excess edit (Lemma 24(4)); for a packing whose members share vertices, the same pair may be a packed pair of another P_3 and is then free, which is the obstruction of [11] and the reason the argument below needs whole components.

Call component j *reducible* if G_j has an optimal solution $S_j \subseteq \binom{W_j}{2}$ such that for all $u, v \in W_j$: (R1) if u, v lie in one cluster of $G_j \triangle S_j$, then $N_G(u) \setminus W_j = N_G(v) \setminus W_j$; (R2) otherwise $N_G(u) \cap N_G(v) \subseteq W_j$. This is Reduction Rule 6.2 of [14] for $H = G_j$.

Lemma 25 (Rule 6.2 at component level)**.** *If component j is reducible with S_j, then $(G, \{G_i\}, k)$ is a YES instance iff $(G \triangle S_j, \{G_i\}_{i \neq j}, k - \tau_j)$ is. In the new instance the graphs G_i ($i \neq j$) are unchanged, still vertex-disjoint and induced, with the same τ_i; the excess ℓ' is unchanged; and for every $i \neq j$ the conditions (R1), (R2), hence reducibility, are unchanged. So one pass over the components, in any order, reaches an instance in which no component is reducible.*

Proof. The equivalence is Lemma 6.3 of [14] (cited; sound for every induced subgraph). Both k and $\sum \tau$ drop by τ_j. The set S_j consists of pairs with both ends in W_j, so no pair incident to a vertex of W_i, $i \neq j$, changes: $G[W_i]$ and all neighbourhoods $N(u)$, $u \in W_i$, are the same before and after, and (R1), (R2) for i refer only to these. □

Lemma 26 (Number of irreducible components)**.** *Let $(G, \{G_j\}_{j \leq r}, k)$ be a YES instance with $\ell' = k - \sum_j \tau_j$ in which no component is reducible. Then $r \leq 2\ell'$.*

Proof. Let S be an optimum of G, so $|S| \leq k$. *Claim:* for every j, (a) $|S_j| \geq \tau_j + 1$, or (b) S contains a pair with exactly one end in W_j (an *external* pair of j). Suppose (a) fails; then $|S_j| = \tau_j$ by Lemma 24(4), so S_j is an optimal solution of G_j, and since j is not reducible, S_j violates (R1) or (R2). In $G \triangle S$ the induced subgraph on W_j is the cluster graph $G_j \triangle S_j$. Case (R1) violated: $u, v \in W_j$ are in one cluster of $G_j \triangle S_j$, so $uv \in E(G \triangle S)$ and, $G \triangle S$ being a cluster graph, $N_{G \triangle S}[u] = N_{G \triangle S}[v]$; but there is $y \notin W_j$ adjacent in G to exactly one of them, so one of uy, vy is in S: an external pair. Case (R2) violated: u, v are in different clusters of $G_j \triangle S_j$, hence non-adjacent in $G \triangle S$, and have a common G-neighbour $y \notin W_j$; if neither uy nor vy were in S, then $u\,y\,v$ would be an induced P_3 of $G \triangle S$. So again an external pair is in S.

Let J_a be the indices satisfying (a) and J_b the others, which satisfy (b). An external pair of j has exactly one end in W_j; it lies in no $\binom{W_i}{2}$ (for $i = j$ because of the end outside, for $i \neq j$ because $W_i \cap W_j = \emptyset$ and one end is in W_j), so it belongs to S_{ext}, and it is external for at most two

indices (the components containing its ends). Hence $|S_{\text{ext}}| \geq |J_b|/2$. Since $|S| = \sum_j |S_j| + |S_{\text{ext}}|$ (disjoint pair sets), $|S| \geq \sum_{j \in J_a} (\tau_j + 1) + \sum_{j \in J_b} \tau_j + |J_b|/2 \geq \sum_j \tau_j + r/2$. With $|S| \leq k$ this reads $\ell' \geq r/2$. □

The bound is on irreducible components, not on components with positive excess $\tau_j > p_j$: a reducible component may have large internal excess, and it is removed by Lemma 25 at no cost in ℓ'.

Proof of Theorem 22. Algorithm. (1) Compute the components of $I(H)$, the optima τ_j by the subset dynamic programme over W_j ($O(3^{|W_j|})$), and ℓ'; if $\ell' < 0$ answer NO (Lemma 24(4)). (2) For each component test reducibility and, if reducible, apply the rule (Lemma 25); the test of [14, Lemma 6.4] needs a constant number of exact cluster-editing computations on subgraphs of G_j plus polynomial time, which the subset programme provides in $O(3^{|W_j|})$ each. (3) Let r^* be the number of remaining components. If $r^* > 2\ell'$, answer NO (Lemma 26, contrapositive). Otherwise the current instance has budget $k^* = \sum_{\text{remaining}} \tau_j + \ell' \leq r^* t_s + \ell' \leq (2t_s + 1)\ell'$. Forget the packing and solve plain Cluster Editing with budget k^* by Böcker's algorithm in $O(1.62^{k^*} + n + m)$.

Correctness follows from Lemma 24 (the instance is of the required form and ℓ' is the excess), Lemma 25 (steps preserve the answer), and Lemma 26.

Time. The components have at least 3 vertices and are disjoint, so there are at most $n/3$ of them and steps (1), (2) take $O(n3^s) + n^{O(1)}$. Step (3) takes $1.62^{k^*} n^{O(1)}$ with $k^* \leq (2t_s+1)\ell' \leq (\binom{s}{2}+1)\ell$, since $2t_s \leq \binom{s}{2}$ and $\ell' \leq \ell$.

Kernel. After step (2) and the checks of step (3), apply the kernelization of Cluster Editing with at most $2k^*$ vertices (cited via [14]) to (G, k^*) and add the empty packing: an equivalent CEaMP instance with empty packing, budget $\leq k^*$ and at most $2k^* \leq (4t_s+2)\ell \leq (s^2 - s + 2)\ell$ vertices. □

Remark 27 (Scope). Theorem 22 is FPT in (ℓ, s); it does not answer the main question, since s is not bounded by a function of w (a path or tree $I(H)$ has $w \leq 1$ and arbitrarily large components), and the exponent $\binom{s}{2}\ell$ is of no practical interest. The argument uses neither pair-maximality nor Section 5. For a perfect matching it covers the smallest non-trivial instance of the case $c = 2$ left open in [11] for FPT in ℓ.

7 Structure of solutions with respect to a packing

This section collects counting lemmas that hold for every packing H (no maximality unless stated) and describe what a solution of size at most $|H| + \ell$ can look like. All proofs use only pair-disjointness, a budget, and counting; none uses optimality of the solution. The lemmas *do not* give an FPT algorithm in $\ell + w$ and no hardness; see Section 7.6 for what they do and do not explain, and Section 11.

Conventions. Here a pair is *hard* if it is not packed (in the language of the previous sections: *unpacked*); a *hard edge* is a hard pair in E and a *hard non-edge* a hard pair outside E. For a packing H let $c = c(H)$ be the maximum number of members of H through a vertex, $D = D(H) = \max_v |Q(v)|$ the maximum number of packed partners, and $w = \text{tw}(I(H))$ (with $w = 0$ if $I(H)$ has no edge). A triple (G, H, ℓ) is a *YES instance* if $\text{OPT}(G) \leq |H| + \ell$. A *family* F of induced P_3's is *hard over* H if the sets $\text{pairs}(P)$, $P \in F$, are pairwise disjoint and disjoint from the packed pairs of H; then $H \cup F$ is a packing with $|H| + |F|$ members.

Lemma 28 (Degrees). *For every vertex v, $|Q(v)| = 2\,|\{P \in H : v \in P\}|$. Hence $D = 2c \leq 2(w+1)$.*

Proof. Each member $P \ni v$ contributes the two packed pairs of P at v, i.e. two partners. Two members through v cannot have a common partner $x \neq v$, since they would then share the two vertices v, x, hence a pair. So the partner sets are disjoint and $|Q(v)| = 2|\{P \ni v\}|$. The inequality $c \leq w + 1$ is stated in Section 2. □

7.1 Maximalisation and the hard-edge lemma

Lemma 29 (Hard families)**.** *Let H be any packing, (G, H, ℓ) a YES instance and F a family hard over H. Then $|F| \leq \ell$. In general* $\mathrm{OPT} \geq |H| + |F|$.

Proof. $H \cup F$ is a modification-disjoint packing of $|H| + |F|$ induced P_3's. Lemma 1, applied to $H \cup F$ and an optimum solution, gives $\mathrm{OPT} \geq |H| + |F|$. In a YES instance $\mathrm{OPT} \leq |H| + \ell$. □

Lemma 30 (MAX: maximalisation)**.** *Let H be any packing and M a maximal family hard over H (inclusion-maximal; it is found greedily in polynomial time). Put $H' = H \cup M$, $\ell' = \ell - |M|$. Then:*

(a) $\mathrm{OPT} \geq |H'|$. *In particular $|M| > \ell$ implies that (G, H, ℓ) is a NO instance.*

(b) $\mathrm{OPT} \leq |H| + \ell$ *iff* ($|M| \leq \ell$ *and* $\mathrm{OPT} \leq |H'| + \ell'$). *So (G, H, ℓ) and (G, H', ℓ') are equivalent whenever $\ell' \geq 0$.*

(c) H' is a packing and is pair-maximal: no induced P_3 of G has all three pairs unpacked with respect to H'. ("Hard" now means hard with respect to H'; the pairs of M have become packed.)

(d) $\mathrm{tw}(I(H')) \leq \mathrm{tw}(I(H)) + |M|$, *hence $w' \leq w + |M| \leq w + \ell$.*

(e) $c' \leq \min\{c + |M|,\ w' + 1\}$ and $D' = 2c' \leq D + 2|M|$.

A tree decomposition of $I(H')$ of width at most $w + |M|$ is obtained in polynomial time from one of $I(H)$.

Proof. (a) is Lemma 29 (first sentence: $\mathrm{OPT} \geq |H| + |M|$). (b) If $\mathrm{OPT} \leq |H| + \ell$ then $|M| \leq \ell$ by (a), and $|H'| + \ell' = |H| + \ell$. Conversely $|M| \leq \ell$ and $\mathrm{OPT} \leq |H'| + \ell' = |H| + \ell$. So (b) is arithmetic once (a) is known.

(c) The members of M have pairwise disjoint pair sets, disjoint from those of H, so H' is a packing. Suppose an induced P_3 Q has all its pairs hard with respect to H'. Then pairs(Q) is disjoint from pairs(H) and from pairs(P) for all $P \in M$, so $M \cup \{Q\}$ is a family hard over H ($Q \notin M$ because the pairs of Q are not in pairs(M)), contradicting maximality of M.

(d) Let $(T, \{B_t\})$ be a tree decomposition of $I(H)$ of width w (if $H = \emptyset$ take a single empty bag). The graph $I(H')$ has node set $H \cup M$ and $I(H')[H] = I(H)$. Put $B'_t = B_t \cup M$. Every node of H still occupies a connected subtree. Every node of M occupies all of T, which is connected. An edge of $I(H')$ with both ends in H is covered as before; an edge with an end $P \in M$ is covered by any bag containing its other end (every node of H lies in some bag; if both ends are in M every bag covers it). Bags have at most $w + 1 + |M|$ nodes, so the width is at most $w + |M|$.

(e) At most $|M|$ members of M pass through a vertex, so $c' \leq c + |M|$; $c' \leq w' + 1$ is Section 2 for H'. By Lemma 28, $D' = 2c' \leq 2c + 2|M| = D + 2|M|$. □

The bound in (d) cannot be improved in general.

Example 31 (Tightness of (d))**.** Let G be the star with centre v and leaves $p, q, a_1, b_1, \ldots, a_m, b_m$, $H = \{(p, v, q)\}$ and $M = \{(a_i, v, b_i) : 1 \leq i \leq m\}$. The members of M are induced P_3's with pairwise disjoint pair sets, disjoint from pairs(H). M is maximal: every induced P_3 of a star is (x, v, y) with leaves x, y, and every pair xv is already packed. $I(H)$ is one node, tw $= 0$; $I(H \cup M) = K_{m+1}$, tw $= m = 0 + |M|$.

After applying Lemma 30, all the other lemmas of this section have to be applied to the primed instance (G, H', ℓ') with parameters $w' \le w + |M|$, $c' \le c + |M|$, $D' \le D + 2|M|$ and $\ell' = \ell - |M|$. This is the same normalisation as Remark 2, with the cost in treewidth made explicit.

Lemma 32 (HE: hard edges close up)**.** *Let H be pair-maximal and let ab, bc be hard edges (a, b, c distinct). Then $ac \in E$ or ac is packed.*

Proof. If $ac \notin E$ and ac were hard, then $a\,b\,c$ would be an induced P_3 with all three pairs unpacked, contradicting pair-maximality. So $ac \in E$ or ac is packed. □

No YES hypothesis is needed. Without pair-maximality the lemma fails ($H = \emptyset$ and any induced P_3). For an arbitrary packing H one applies it to $H' = H \cup M$ of Lemma 30(c), with "hard" meaning hard with respect to H'.

7.2 Hubs

Lemma 33 (HUB$^+$)**.** *Let H be any packing, (G, H, ℓ) a YES instance, u a vertex and $N_h(u) = \{x : ux \in E,\ ux \text{ not packed}\}$. Let Γ_u be the graph on $N_h(u)$ whose edges are the pairs $\{x, y\}$ with $xy \notin E$ and xy not packed (hard non-edges). Then:*

(a) $\nu(\Gamma_u) \le \ell$ (matching number);

(b) Γ_u has a vertex cover T with $|T| \le 2\ell$, and every pair of $N_h(u) \setminus T$ is an edge of G or is packed;

(c) $\alpha(G[N_h(u)]) \le 2\ell + c + 1$ where $c = c(H)$.

Proof. (a) Let $x_1y_1, \ldots, x_ty_t$ be a matching of Γ_u and $P_i = (x_i, u, y_i)$. Since $ux_i, uy_i \in E$ and $x_iy_i \notin E$, P_i is an induced P_3. Its three pairs ux_i, uy_i, x_iy_i are hard. The $2t$ vertices x_i, y_i are distinct and different from u, so the pairs ux_i, uy_i ($i \le t$) are pairwise distinct, and the pairs x_iy_i are pairwise distinct and do not contain u. So $\{P_1, \ldots, P_t\}$ is hard over H and $t \le \ell$ by Lemma 29.

(b) The vertex set of a maximum matching of Γ_u is a vertex cover (by maximality) of size $2\nu \le 2\ell$. A pair of $N_h(u) \setminus T$ is not an edge of Γ_u, i.e. it is an edge of G or packed.

(c) Let J be independent in $G[N_h(u)]$. By (b), every pair of $J \setminus T$ is a non-edge of G and not an edge of Γ_u, hence packed. Put $J' = J \setminus T$. A packed pair of J' lies in a member P of H and is a non-edge, so it is the pair of the two ends of P (the other two pairs of P are edges); in particular $|V(P) \cap J'| \le 2$. If $J' \ne \emptyset$ fix $x \in J'$; every $y \in J' \setminus \{x\}$ gives a member $P_y \ni x, y$, and $P_y \ne P_{y'}$ for $y \ne y'$ because $|V(P_y) \cap J'| \le 2$. So $|J'| - 1 \le c$. Hence $|J| \le |T| + |J'| \le 2\ell + c + 1$. □

Corollary 34 (HUB, hard-independent sets)**.** *In the setting of Lemma 33, if $X \subseteq N_h(u)$ and every pair of X is a hard non-edge, then $|X| \le 2\ell + 1$.*

Proof. X is a clique of Γ_u, so it contains a matching of size $\lfloor |X|/2 \rfloor$, and $\lfloor |X|/2 \rfloor \le \ell$ by Lemma 33(a). □

Remark 35 (HUB$^+$ versus an independence-number bound). (1) Bounding $\alpha(G[N_h(u)])$ through the corollary (greedy choice of a subset of an independent set that is also free of packed pairs) gives only $(c+1)(2\ell+1) = (D/2+1)(2\ell+1)$, which is weaker than (c). (2) The matching form (a) is the relevant statement for Vertex Cover on the non-adjacency graph of $N(u)$: let $N_h(u)$ be two disjoint cliques of G of size m with no packed pair between them. Then $\alpha = 2$, but (a_i, u, b_i), a_i, b_i from different cliques, are m pair-disjoint induced P_3's, and $m \le \ell$ in a YES instance. Large independent sets are not the obstruction; large matchings of Γ_u are. (3) For a general H the lemma is applied to H' of Lemma 30 with c', ℓ'.

7.3 Pieces and small merges

Fix a packing H and a solution S ($G \triangle S$ is a cluster graph) with $|S| \le |H| + \ell$. Let $U = S \setminus \bigcup_i \text{pairs}(P_i)$ be the unpacked part and $u = |U|$; by Lemma 1, $u \le |S| - |H| \le \ell$. Put $F = U \cap E$ (deleted hard edges) and $I = U \setminus E$ (inserted hard pairs). The *pieces of* S are the connected components of the graph (V, K), $K = \{\text{hard edges of } G\} \setminus F$ (kept hard edges); isolated vertices are singleton pieces. *Pieces depend on S, not only on (G, H).* Every edge of K is an edge of $G \triangle S$, so every piece lies inside one cluster of $G \triangle S$.

Lemma 36 (SM: small merge)**.** *Let S, u, I, pieces as above, $D \ge \max_v |Q(v)|$ (e.g. $D = D(H)$), and let C be a cluster of $G \triangle S$.*

(i) If $A \ne B$ are pieces contained in C, then every pair ab ($a \in A, b \in B$) is packed or belongs to I.

(ii) For such A, B with $|A| \ge |B|$: $|B|\,(|A| - D) \le |I \cap (A \times B)| \le u$, hence $|A| \le D + u \le D + \ell$.

(iii) If C is the union of at least two pieces, then $|C| \le \max\{D + u + 1,\ 2D,\ 2u\}$.

Proof. (i) Since $a, b \in C$, ab is an edge of $G \triangle S$, i.e. $ab \in E$ xor $ab \in S$. Suppose ab is hard. If $ab \in E$ then $ab \notin S$; as $F \subseteq S$, $ab \notin F$, so $ab \in K$ and a, b lie in one piece, a contradiction. So $ab \notin E$, hence $ab \in S$, and as ab is hard, $ab \in U \setminus E = I$.

(ii) Each vertex of B has at most D packed partners, so at most $D|B|$ pairs of $A \times B$ are packed. By (i) all other pairs of $A \times B$, at least $|A||B| - D|B|$, lie in I, and $|I| \le |U| = u$. If $|A| \ge D + u + 1$ then $|B|(|A| - D) \ge u + 1$ because $|B| \ge 1$, a contradiction; so $|A| \le D + u$.

(iii) Let A be a largest piece in C, $a = |A|$ and $r = |C| - a \ge 1$. By (i), every pair $\{x, y\}$ with $x, y \in C$ in different pieces is packed or in I.

Case $a > D$. Each $x \in C \setminus A$ has at most D packed partners, so at least $a - D \ge 1$ vertices $y \in A$ give a pair xy in I. These pairs are distinct for distinct (x, y), so $r(a - D) \le |I| \le u$. Then $r \le u$ and $a \le D + u/r$, so $|C| = a + r \le D + u/r + r$. The function $r \mapsto r + u/r$ is convex on $[1, u]$ and takes the value $u + 1$ at both ends, so $|C| \le D + u + 1$. (The case does not occur if $u = 0$.)

Case $a \le D$. All pieces of C have at most D vertices. Each $x \in C$ has at least $|C| - D$ vertices of C outside its piece, at most D of which are packed partners; so x lies in at least $|C| - 2D$ pairs $\{x, y\}$ of I with $y \in C$. Summing over x counts each pair twice: $2|I| \ge |C|(|C| - 2D)$. If $|C| > 2D$, then $|C| - 2D \ge 1$ and $|C| \le 2|I| \le 2u$. So $|C| \le \max\{2D, 2u\}$. □

Remark 37 (Qualifiers and relation to the original bound). (1) The qualifier "pieces sharing a cluster" is essential: a single piece may be an entire cluster of unbounded size, and a statement "pieces have size at most $D + \ell$" is false. (2) The bound (iii) implies the weaker bound $|C| \le \max\{2D + \ell, 2\ell\}$ with which the lemma first appeared in our notes: $2u \le 2\ell$ and $2D \le 2D + \ell$ are clear; $D + u + 1 \le 2D + \ell$ if $D \ge 1$; if $D = 0$ and $u \ge 1$ then $u + 1 \le 2u \le 2\ell$; and if $D = u = 0$ no cluster with two pieces exists (the two cases of the proof are both impossible). (3) The bound (iii) is attained in some small instances of our tests (slack 0), for instance two singleton pieces and one inserted pair; we do not claim it is tight in general. (4) Lemma 36 says nothing about the *degrees* in a graph of pieces; it bounds the sizes of pieces that share a cluster and the number of cross pairs that are not packed.

7.4 The soft graph has bounded treewidth

Let $W = (V, \{\text{packed pairs}\})$ be the *soft graph*.

Proposition 38 (Treewidth of the soft graph)**.** $\text{tw}(W) \le 3(w + 1) - 1 = 3w + 2$ *if $H \ne \emptyset$; W is edgeless if $H = \emptyset$. The bound is attained for $w = 0$ (H a single P_3, W a triangle).*

Proof. Let $(T, \{B_t\})$ be a tree decomposition of $I(H)$ of width w and $T_P = \{t : P \in B_t\}$, a subtree of T for each $P \in H$. Put $B'_t = \bigcup_{P \in B_t} V(P)$, so $|B'_t| \le 3(w+1)$. For every vertex $x \notin V(H)$ add a new leaf bag $\{x\}$ to an arbitrary node. *Edges:* a packed pair lies in $V(P)$ for some $P \in H$, and $P \in B_t$ for some t, so the pair lies in B'_t. *Connectivity:* for $x \in V(H)$ the nodes t with $x \in B'_t$ form $\bigcup_{P \ni x} T_P$. The members through x pairwise share x, so they are pairwise adjacent in $I(H)$ and, the edge being covered by a bag, their subtrees T_P pairwise intersect. A union of connected subtrees that pairwise intersect is connected (two points of the union lie in subtrees $T_P, T_{P'}$ that meet; join them through a common node). The new leaf bags are single bags of size 1. So this is a tree decomposition of W of width at most $3(w+1) - 1$. □

Computationally, for $w \ge 1$ the largest value of $\mathrm{tw}(W)$ that we saw on small instances is much smaller than $3w + 2$; we do not claim the bound is tight beyond $w = 0$.

7.5 Computer verification

All statements of this section were checked by brute force with code written independently of the other scripts (`code/w5_packing_lemmas.py`, all set partitions, $n \le 8$, 4000 random graphs: random, noisy clusters, hubs; for each a random partial packing H and its maximalisation $H \cup M$): Lemma 28 ($D = 2c$), Lemma 30(a)–(e), Example 31 for $m \le 5$, Lemma 32, Lemma 33 and Corollary 34 for every vertex and every ℓ with $\mathrm{OPT} \le |H| + \ell \le |H| + 3$, Lemma 36 on every solution of cost at most $|H| + 3$ (about $5.7 \cdot 10^5$ clusters with two or more pieces, including slack 0 in (iii) on some cases), and Proposition 38 (exact treewidth). No violation; the variants $\alpha \le 2\ell + c$ and $|C| \le \max\{D + u, 2D, 2u\}$ are falsified by the same script, so the checks are not vacuous. This is evidence for the statements; the proofs above are the justification.

7.6 Why simple hardness encodings fail

This subsection is a heuristic discussion, not a theorem. It explains why three natural ways of building W[1]-hardness for $\ell + w$ do not work as stated; it does not exclude other gadgets and it does not prove fixed-parameter tractability, which remains open (Open Problem 1). A reduction from a problem with parameter k must keep $\ell + w \le g(k)$; grids of side k inside $I(H)$ are therefore allowed, but structures that grow with the instance size are not.

Hub of deletions. A hub u is adjacent to vertices $x_1, \ldots, x_m$ and the instance asks to keep a subset of $N(u)$ forming a clique and delete the rest, i.e. Vertex Cover on the non-adjacency graph of $N(u)$. Lemma 33 shows what survives in a YES instance: the graph of *hard* non-adjacent pairs inside $N_h(u)$ has a matching of size at most ℓ and a vertex cover of size at most 2ℓ, so at most 2ℓ hub neighbours carry hard conflicts; the other non-adjacencies are adjacencies or packed pairs, and a vertex has at most $2c$ of the latter. The relevant obstruction is a large matching (Remark 35(2)), not a large independent set, and any such matching immediately forces a NO answer. Non-adjacencies realised by packed pairs are local, with degree at most $2c \le 2(w+1)$ in the soft graph. This is why Vertex Cover through a hub cannot encode an unbounded source graph; it does not address hubs whose conflicts are realised by packed pairs; a vertex has at most $2c$ of these, so many of them force large c, hence large w.

Hub of insertions. A hub u with classes $V_1, \ldots, V_k$ chooses, by inserting uv_{i,j_i}, one vertex per class; the selected vertices land in one cluster C and compatibility is encoded by adjacency of the chosen vertices. By Lemma 36(i), two selected vertices in different pieces of C form a packed pair

or an inserted pair (at most ℓ of these); adjacency by a hard edge that is not deleted puts them in the same piece, and Lemma 36(ii) bounds the size of merged pieces by $D + \ell$. So the compatibility relation between selected vertices in different pieces can only be realised by packed pairs, which form the soft graph, of treewidth at most $3w + 2$ (Proposition 38), or by a bounded number of inserted pairs. The caveats: Lemma 36 bounds sizes of pieces, not the degree of a compatibility graph; at most ℓ hard edges may be deleted, which a gadget could exploit; and pieces depend on the chosen solution. The argument shows that the hard graph cannot serve as a channel for $[n]$-valued information between selected vertices, nothing more.

Grid tiling. A tile vertex t_{ab} is joined by hard edges to a row vertex x_a and a column vertex y_b ($x_a y_b$ a non-edge). The triples (x_a, t_{ab}, y_b) are induced P_3's; if their pairs are hard they are pairwise pair-disjoint (their pairs are $x_a t_{ab}$, $t_{ab} y_b$, $x_a y_b$, all different for different (a, b)), so by Lemma 29 at most ℓ of them can be hard, whereas a grid-tiling encoding needs about n^2 tiles and n is not bounded by $g(k)$. So all but at most ℓ of them are packed; then some row vertex x_a lies in at least $n - \ell$ members (if every row has n tiles), $c \geq n - \ell$, hence $w \geq n - \ell - 1$, again unbounded in k. Mixed and replicated gadgets (several copies of row and column vertices, tiles realised by chains) were not excluded by a proof; the examination in our notes found no variant that carries information on $[n]$ across a tree decomposition of bounded width, which is evidence only.

What remains open. The lemmas above make every "simple" source of unbounded information in $I(H)$-width-bounded instances expensive. They do not bound the candidate positions of inserted pairs between pieces. Sections 8 and 9 go further: the inserted pairs of an optimal solution are local for type T1, in particular for merges of pieces with no soft pair between them, but not in general (Proposition 45), and given one endpoint of an edited hard pair the other has a bounded candidate set (Corollary 54). The FPT question for $\ell + w$ stays an open problem.

8 Locality of inserted pairs in optimal solutions

Section 7 bounds the *number* of unpacked edits and the *size* of merged pieces, but not the number of candidate *positions* of an unpacked edit; a positive answer to Open Problem 1 would need such a bound. This section studies the inserted unpacked pairs of an *optimal* solution. The main tool is an elementary diameter bound for clusters of optimal solutions (Lemma 39), which, combined with pair-maximality, classifies every inserted pair into one of two types (Theorem 40). Pairs of the first type are local in a decomposition of $I(H)$ (Corollary 41, Proposition 43); we then show that the second type, which includes the generic merge of two pieces, is *not* local in the same sense (Proposition 45). **Nothing in this section gives an FPT algorithm in $\ell + w$ or a hardness result**; Section 8.6 states exactly what is left.

Setting. Throughout, H is *pair-maximal* (for an arbitrary packing apply Lemma 30 first and read all parameters for H'). A pair is *soft* if it is packed and *hard* otherwise; $Q(v)$ is the set of soft partners of v, $D \geq \max_v |Q(v)|$ and, for $D = D(H) = 2c$, $|Q(v)| \leq D$ (Lemma 28). S is an *optimal* solution, $|S| = \mathrm{OPT}(G)$, C is a cluster of $G \triangle S$, $U = S \setminus \bigcup_i \mathrm{pairs}(P_i)$, $u = |U|$, $F = U \cap E$, $I = U \setminus E$, and the *pieces* are the components of the graph of kept hard edges, as in Section 7. An *inserted pair* is a pair of I: a hard non-edge of G inside one cluster. A hard edge of G inside a cluster is an edge of $G \triangle S$ and not in S, hence kept; so its ends lie in one piece. Cross pairs between two pieces inside one cluster are soft or inserted (Lemma 36(i)).

8.1 Clusters of optimal solutions have diameter at most two

Lemma 39 (Half-degree and diameter)**.** *Let S be an optimal solution, C a cluster of $G \triangle S$ and $v \in C$. Put $d_C(v) = |N_G(v) \cap C|$.*

(a) $d_C(v) \geq (|C| - 1)/2$.

(b) If $a, b \in C$ and $ab \notin E$, then some $z \in C \setminus \{a, b\}$ has $az, zb \in E$.

Proof. (a) Let $S' = S \triangle \{vx : x \in C \setminus \{v\}\}$. Then $G \triangle S'$ is $G \triangle S$ with v cut off from the rest of C, a cluster graph. For $x \in C \setminus \{v\}$ with $vx \in E$ the pair $vx \notin S$ and $vx \in S'$: this adds $d_C(v)$ pairs. For $x \in C \setminus \{v\}$ with $vx \notin E$ the pair is in S and not in S': this removes $|C| - 1 - d_C(v)$ pairs. All other pairs are unchanged. So $|S'| = |S| + 2d_C(v) - (|C| - 1)$, and $|S'| \geq |S|$ by optimality.

(b) $N_G(a) \cap C$ and $N_G(b) \cap C$ are subsets of $C \setminus \{a, b\}$ (as $a \notin N(a)$ and $ab \notin E$), a set with $|C| - 2$ elements, while $d_C(a) + d_C(b) \geq |C| - 1$ by (a). So the two sets intersect. □

Lemma 39 uses neither H nor ℓ; only local optimality under single-vertex moves is used.

8.2 The T1/T2 dichotomy

Theorem 40 (T1/T2)**.** *Let H be pair-maximal, S an optimal solution and ab an inserted pair, $a, b \in C$. Then there is $z \in C \setminus \{a, b\}$ with $az, zb \in E$, and exactly one of the following holds.*

(T1) az and zb are both soft. Then they are owned by two distinct *members $P_a \neq P_b$ of H, and $z \in V(P_a) \cap V(P_b)$.*

(T2) Exactly one of az, zb is hard, say az (otherwise swap a, b). Then az is a kept hard edge, so a and z lie in one piece, and zb is soft.

Proof. Take z from Lemma 39(b). If az and zb were both hard, then $a\,z\,b$ would be an induced P_3 with all three pairs hard (ab is hard by assumption), contradicting pair-maximality. If both are soft, let P_a, P_b own az, zb. If $P_a = P_b$ then $a, b \in V(P_a)$ and ab is packed, contradicting that ab is hard; so $P_a \neq P_b$, and both contain z. If az is hard, it is a hard edge inside C, hence kept. □

Corollary 41 (Locality of soft-free merges)**.** *Let H be pair-maximal, S optimal, and let $A \neq B$ be pieces inside one cluster C such that* no soft pair joins A and B. *Then for every $(a, b) \in A \times B$:*

(i) ab is an inserted pair;

(ii) there is $z \in C \setminus (A \cup B)$ with az, zb soft, owned by distinct members $P_a \neq P_b$ with $z \in V(P_a) \cap V(P_b)$;

(iii) given a, the number of possible partners b in any such pair is at most $D(D-2) = 4c(c-1)$, where $D = D(H) = 2c$.

In particular a and b are at distance exactly 2 in the soft graph W, and $A \cup B$ lies in the ball of radius 4 of W around any of its vertices.

Proof. (i) The cross pair ab is not soft by hypothesis, so it is hard. A hard edge inside C would be kept and would put a, b into one piece. So ab is a hard non-edge in a cluster, i.e. inserted.

(ii) Apply Theorem 40 to ab. In case (T2), say az is hard (the other case is symmetric); then z lies in the piece A of a. The pair zb is an edge of G and joins A to B: if it were hard it would be kept and z, b would share a piece, contradicting $A \neq B$; if soft it contradicts the hypothesis. So

(T1) holds. If $z \in A$, then zb (soft) is a soft pair between A and B; similarly for $z \in B$ and az. So $z \notin A \cup B$.

(iii) Fix a. By (ii) $z \in Q(a)$: at most D choices. For such z, let P_a own az. The vertex b satisfies bz soft owned by $P_b \neq P_a$, and $b \notin V(P_a)$ (otherwise $ab \in \text{pairs}(P_a)$ is packed). Now $Q(z) \cap V(P_a) = V(P_a) \setminus \{z\}$ has two elements, so $b \in Q(z) \setminus V(P_a)$, a set of at most $|Q(z)| - 2 \leq D - 2$ vertices. Altogether at most $D(D-2) = 2c(2c-2) = 4c(c-1)$ partners. (For $c = 1$ the count is 0, as it must be: (T1) needs two members through z.)

The distance statement: ab is not soft and a, z, b is a soft path; so $d_W(a, b) = 2$. Every $a' \in A$ has $d_W(a', b) = 2$ for a fixed $b \in B$, and every $b' \in B$ has $d_W(b', a') = 2$; hence all of $A \cup B$ is within distance 4 of b (and of any other vertex of $A \cup B$, by symmetry). □

Remark 42 (Scope; what the corollary does not say). By Lemma 36, if no soft pair joins A and B then all $|A||B|$ cross pairs are inserted, so $|A||B| \leq u \leq \ell$. The corollary covers *only* merges without a soft pair between the pieces. Merges in which some cross pairs are soft (the generic case when the pieces are neighbours in the soft graph) are not covered, and Proposition 45 shows they can genuinely be of type (T2). Note also that in terms of soft distance the bound in (iii) is the number of vertices at soft distance exactly 2 with the stated restrictions; the content of (ii) is that in the soft-free case *every* cross pair is at soft distance 2.

Proposition 43 (Bag locality of type (T1) pairs)**.** *Let $(T, \{X_t\})$ be a tree decomposition of $I(H)$ of width w, let $c = c(H)$, and for a set Y of nodes of $I(H)$ let $N[Y]$ be the set of nodes at distance at most 1 from Y (a set of* members *of H).*

(a) If ab is an inserted pair of type (T1) with members P_a, P_b as in Theorem 40, then P_a, P_b are adjacent in $I(H)$, so some bag X_t contains both and $a, b \in \bigcup_{P \in X_t} V(P)$, a set of at most $3(w+1)$ vertices. Moreover the member through z and every member through a or b lies in $N[X_t]$.

(b) $(T, \{N[X_t]\})$ is a tree decomposition of the square $I(H)^2$ of width at most $(w+1)(3c-2)-1 \leq (w+1)(3w+1)-1$.

Proof. (a) P_a, P_b share z, so they are adjacent in $I(H)$ and, the edge being covered, lie in a common bag. Every member through a contains $a \in V(P_a)$, hence meets $P_a \in X_t$ and lies in $N[X_t]$; similarly for b and z.

(b) *Size.* Each member has 3 vertices, each lying in at most c members, so $I(H)$ has maximum degree at most $3(c-1)$, $|N[X_t]| \leq (w+1)(1+3(c-1)) = (w+1)(3c-2)$, and $c \leq w+1$. *Connectivity.* For a node v let $T_v = \{t : v \in X_t\}$ (a subtree). The nodes t with $v \in N[X_t]$ form $\bigcup_{y \in N[v]} T_y$. Each T_y, $y \in N(v)$, meets T_v because the edge yv lies in some bag; a union of subtrees that all meet the connected set T_v is connected. *Edges.* Let yv be an edge of $I(H)^2$. If y, v are adjacent in $I(H)$, any $t \in T_y$ has $y \in X_t$ and $v \in N[y]$. If they have a common neighbour m, any $t \in T_m$ has $y, v \in N[m] \subseteq N[X_t]$. □

A type (T1) pointer is therefore a pair of vertices inside one bag of a decomposition of bounded width, and, given one end, it has at most $4c(c-1)$ candidate partners in the soft-free case. No identity problem between far bags arises for these pointers.

8.3 Near-twin agreement and anchored candidates

Lemma 44 (Near-twin agreement)**.** *Let S be any solution (optimal or not), $u = |U|$, and let $z \neq b$ lie in one cluster of $G \triangle S$. Then*

$$A(z, b) = \big\{a \notin \{z, b\} : \ az, ab \text{ both hard}, \ [az \in E] \neq [ab \in E]\big\}$$

has at most u elements. In particular, if zb is soft, the set $M(z,b) = N_h(z) \setminus (N[b] \cup Q(b))$, where $N_h(z)$ is the set of hard neighbours of z, satisfies $|M(z,b)| \le u \le |S| - |H|$ and is computable from G and H alone.

Proof. Let $a \in A(z,b)$. Exactly one of az, ab is a hard edge and the other a hard non-edge. If a lies in the cluster of z, b, the hard non-edge is a hard non-edge inside a cluster, so it is in U. If a lies in another cluster, the hard edge joins two different clusters, so it is deleted and in U. Assign to a this pair. Its ends are a and one of z, b; since $a \notin \{z, b\}$, the map $a \mapsto$ pair is injective. So $|A(z,b)| \le u$. For the second statement, $a \in M(z,b)$ means $a \neq z$, az a hard edge, $a \notin N[b]$ (so $a \neq b$ and $ab \notin E$) and ab not soft. Thus $a \in A(z,b)$. □

Lemma 44 is the sense in which a far end of a type (T2) pointer is determined by its anchor: in (T2) with az hard and zb soft, $a \in M(z,b)$, a set of at most ℓ vertices determined by the *soft* pair (z,b). It does not say where a is relative to z in $I(H)$: a is a hard neighbour of z.

8.4 Merge pointers need not be bag-local

Proposition 45 (Computer-assisted examples)**.** *Each of the following instances (graph G on vertices $0, \dots, n-1$ given by its edge list, H given by triples (x,y,z) with centre y) has a pair-maximal modification-disjoint packing H and a unique optimal solution, found by enumerating all set partitions of the vertex set.*

(a) $n = 7$, $E = \{03, 04, 12, 13, 14, 15, 23, 35, 36\}$, $H = \{(4,1,5), (0,3,2), (1,3,6)\}$. $\mathrm{OPT} = 4 = |H| + 1$*; the clusters are* $\{0,4\}, \{1,2,3,5\}, \{6\}$*. The pieces of* $\{1,2,3,5\}$ *are* $\{1,2\}$ *and* $\{3,5\}$ *(kept hard edges* 12 *and* 35*); the cross pairs* $13, 15, 23$ *are soft and* 25 *is inserted.* $Q(2) = \{0,3\}$ *and* $Q(5) = \{4,1\}$ *are disjoint, so the pair* 25*, whose ends lie in two different pieces, has no common soft neighbour: it is of type (T2) and not of type (T1).*

(b) $n = 8$, $E = \{02, 06, 14, 16, 17, 23, 25, 37, 45, 46, 47, 57\}$, $H = \{(0,6,4), (3,7,4), (2,5,4), (0,2,3), (6,1,7)\}$. $\mathrm{OPT} = 6 = |H| + 1$*; the clusters are* $\{0,6\}, \{1,4,5,7\}, \{2,3\}$*; the only unpacked pair of the solution is the inserted pair* 15*; the pieces are* $\{1,4\}$ *and* $\{5,7\}$*, the cross pairs* $17, 45, 47$ *are soft,* $Q(1) = \{6,7\}$*,* $Q(5) = \{2,4\}$*.*

(c) $n = 6$, $E = \{02, 03, 04, 05, 12, 13, 15, 25, 35, 45\}$, $H = \{(3,5,4), (0,2,1)\}$*. The unique optimum has clusters* $\{0,1,2,3,5\}, \{4\}$ *and* $\mathrm{OPT} = 4$*; it contains the inserted pair* 23 *(type (T2):* 30 *hard and kept,* 02 *soft), but* 2 *and* 3 *lie in the* same *piece (the kept hard edges* $03, 05, 13, 15, 25$ *connect* $\{0,1,2,3,5\}$*).*

Consequently "every inserted pair with both ends in $V(H)$ has a common soft neighbour" is false, and the statement of Corollary 41 cannot be extended to merges of two pieces that are joined by a soft pair: (a) and (b) are such merges, with inserted pairs of type (T2).

Proof. Direct enumeration (all 877, 4140 and 203 set partitions of 7, 8 and 6 vertices respectively) of the cost $|\{ab : (ab \in E) \neq (a,b \text{ in one block})\}|$, together with a check that each triple of H is an induced P_3, that the triples have pairwise disjoint pair sets and that no induced P_3 of G has all pairs unpacked. Examples (a) and (c) were found by an independent search in the audit of our notes, (b) in the notes themselves; all three were re-checked for this paper. The structural claims (pieces, soft pairs, Q) are read off from the lists above. □

Remark 46 (Size of the examples). An exhaustive search over all labelled graphs and all pair-maximal packings with $n \le 5$ found no instance in which every optimum has an inserted pair with

both ends in $V(H)$ and no common soft neighbour (computer evidence of the audit; not a theorem). Example (c) ($n = 6$) is intra-piece; the smallest merge-type example found has $n = 7$ (none in 7,509 random draws at $n = 6$, 9 in 7,815 at $n = 7$).

Remark 47 (Far type (T2) evidence is intra-piece). In the notes' experiments with H equal to three vertex-disjoint P_3's ($c = 1$, $n = 9$) there are instances in which every optimum contains an inserted pair between vertices of different members of H, hence at infinite distance in $I(H)$ (computer observation, reproduced in the audit). In the instance examined in the audit (inserted pair 58 in every one of its six optima) the two ends lie in the *same piece*: the cluster is $\{4, 5, 7, 8\}$ with kept hard edges $47, 48, 57$. The pair is determined by hard connectivity, i.e. by G, and is not a free pointer between two pieces. In the audit's search ($n = 9$, three disjoint P_3's, 324 valid random instances) no instance had a merge-type inserted pair between different members in every optimum. So this evidence shows non-locality only for intra-piece insertions; *whether merge pointers between pieces can be forced at unbounded $I(H)$-distance is open* (neither proved nor refuted; larger instances than $n = 9$, $c = 1$ would be needed).

8.5 Computer evidence

For every set partition of $n = 6, \ldots, 9$ vertices of about 1,900 pair-maximal instances (random, noisy clusters, hubs, sparse, algebraic gadgets, random and overlapping maximal packings), and for $\approx$ 5,700 optimal partitions found by an ILP solver for $n = 11, 12, 13$, the notes' checker tested Theorem 40, Corollary 41 and Lemma 44: no violations among 7,082 + 5,700 optimal partitions. As a control, on partitions of cost OPT + 1 and OPT + 2 the conclusions of Lemma 39, Theorem 40 and Corollary 41 fail thousands of times, while the purely counting Lemma 44 never fails, as it should. An independently written checker (audit) confirmed Lemma 39 and Theorem 40 on all optimal partitions of random pair-maximal instances for $n = 6, 7, 8$ (744 + 517 + 137 optimal partitions), Lemma 44 on 511,560 pairs (z, b) over all partitions of $n = 7$, and the decomposition of Proposition 43(b) on 4,000 random triple systems. This is evidence; the proofs above are the justification.

8.6 What remains

Every inserted pair of an optimal solution is of type (T1) or (T2) (Theorem 40). What is established is the following.

- Type (T1) pairs, in particular all cross pairs of soft-free merges, are bag-local with an explicit bound on the number of partners (Corollary 41, Proposition 43).
- Merges of two pieces joined by a soft pair can be of type (T2) (Proposition 45), so the closed statement is *soft-free merges only*; the claim "tiny pointers are local" is not established in general.

What remains open:

(R1) *Type (T2) pointers.* By Lemma 44 the far end a of a (T2) pair is one of at most ℓ vertices fixed by the soft anchor (z, b), but a is at hard distance 1 from z, so it can be far from z in $I(H)$ (the hard edge az may join vertices of far-apart members). Whether the interaction of these far ends with the rest of the solution can be handled inside a bounded-width decomposition is open. (A conjecture that (T2) pointers can be absorbed into the base CSP of Section 3–4 by a per-anchor unary constraint plus at most ℓ extra pairs is untested; we state it only as a possibility.)

(R2) *Deleted hard edges.* A deleted hard edge between far members is the same kind of object (a hard edge used as a tunnel); this section says nothing about it. The example of Section 9 (Proposition 61) shows that free hard edges between far vertices cannot be ignored when one tries to build a single decomposition for all purposes.

(R3) *The missing reduction lemma* (Conjecture 64 in Section 9).

Even a complete solution of (R1) and (R2) would not decide Open Problem 1 unless it also handles the position of $\leq \ell$ bounded pieces, which Section 9 reduces to an n^ℓ guess.

Remark 48 (Heuristic, not a theorem: why a hardness proof must use (T2) or deletions). Since (T1) pairs are bag-local, any W[1]-hardness reduction for $\ell+w$ would have to transport information through (T2) pairs or deleted hard edges. A kept hard edge az with a far from z can force the final status of the soft pairs at a to agree with those at z, i.e. it realises *equality* of colours between far positions; Grid Tiling needs exactly equalities between neighbouring cells together with local membership constraints. We could not complete such a construction: an inactive anchor must cost nothing, $|M(z,b)| \leq \ell$ forces each anchor's far end to be unique (so the colour would have to be encoded by *which* anchor is active), and the independence-number bound of Lemma 33 must not be violated by many anchors that share hard neighbours. This is an unverified idea and is listed only to say where a hardness proof would have to look.

9 The cluster-identity route: bounded candidates, an n^ℓ algorithm and an FPT algorithm in (ℓ, D, ω)

This section pursues the following route to fixed-parameter tractability in $\ell + w$: describe the clusters of an optimal solution by a normal form (Lemmas 49–53), show that one endpoint of an unpacked edit determines a bounded candidate set for the other (Corollary 54), and run a dynamic program that either guesses the positions of the edits (Theorem 57) or follows the clusters directly (Theorem 59). The route yields two proved results, a halving of the exponent in Theorem 6 and an FPT algorithm for a stronger parameter ω, and **it does not yield FPT in** $\ell + w$: ω is not bounded by any function of (ℓ, w) (Proposition 61), and the reduction that would repair this is missing (Conjecture 64). The question "FPT in (ℓ, w)?" remains **open**.

Setting and notation. H is pair-maximal; for an arbitrary packing apply Lemma 30 and read w, D, ℓ as the primed values $w' \leq w + |M|$, $D' \leq D + 2|M|$, $\ell' = \ell - |M|$ (Remark 58 below). "Soft" means packed, "hard" unpacked; $Q(v)$ is the set of soft partners of v and D is any number with $D \geq \max_v |Q(v)|$ (for example $D = D(H) = 2c$). $N_h(v)$ is the set of hard neighbours and $d_h(v) = |N_h(v)|$. A *solution* is a set S of pairs with $G \triangle S$ a cluster graph; its *clusters* are the cliques of $G \triangle S$, and every partition π of V is the cluster partition of exactly one solution, namely the set of pairs ab with $[a, b \text{ in one block}] \neq [ab \in E]$; its size is the *cost* of π. As in Section 7, U is the set of *edited hard pairs* of S (hard pairs in S), with $u = |U| \leq |S| - |H|$ (Lemma 1); del_x and ins_x count the deleted hard edges (hard edges across clusters) and inserted hard non-edges (hard non-edges inside clusters) of U at the vertex x. A *hard component* is a connected component of the graph $(V, \text{hard edges of } G)$; it is *clean* if all its non-adjacent pairs are soft. We say (G, H, ℓ) is a YES instance if $\mathrm{OPT} \leq |H| + \ell$, and then every solution of size $\leq |H| + \ell$ has $u \leq \ell$.

9.1 Cost decomposition

Lemma 49 (CP: cost decomposition)**.** *Let π be a partition of V and $f(C) = \binom{|C|}{2} - 2e(C)$, where $e(C)$ is the number of edges of G inside C.*

(a) $\mathrm{cost}(\pi) = |E| + \sum_{C\in\pi} f(C)$.

(b) $\mathrm{cost}(\pi) = \sum_i c_i + |U|$, *where $c_i \geq 1$ is the number of edited pairs among the three pairs of P_i (it depends only on π restricted to $V(P_i)$). The excess is $\sum_i (c_i - 1) + |U|$.*

(c) Let (T, β) be a tree decomposition of the soft graph W (Proposition 38) and call a hard pair local *if it lies in a common bag. Then $\mathrm{cost}(\pi) = \sum_t \mathrm{loc}_t + \mathrm{far}$, where each edited soft pair and each edited local hard pair is charged to one bag containing it, loc_t depends only on $\pi|_{\beta(t)}$, and* far *is the number of edited non-local hard pairs, so* $\mathrm{far} \leq |U|$.

Proof. (a) The cost counts non-edges inside blocks, $\sum_C [\binom{|C|}{2} - e(C)]$, plus edges across blocks, $|E| - \sum_C e(C)$. (b) The pair sets $\mathrm{pairs}(P_i)$ are disjoint and are exactly the soft pairs, and $c_i \geq 1$ since otherwise the triple $V(P_i)$ would stay an induced P_3 of $G \triangle S$ (Lemma 1). (c) is bookkeeping: all edited pairs are soft, or hard and local, or hard and non-local; soft pairs always lie in a bag. □

(The $\ell = 0$ CSP of Theorem 5 is the local part; all non-locality is in far.)

9.2 Structure of the hard pairs

Lemma 32 (HE) is the first statement used below. We add its companion.

Lemma 50 (H2)**.** *Let H be pair-maximal and uv a hard non-edge. Then $N_h(u) \cap N_h(v) = \emptyset$.*

Proof. If $z \in N_h(u) \cap N_h(v)$, then $u\,z\,v$ is an induced P_3 ($uz, zv \in E$, $uv \notin E$) all of whose pairs are hard, contradicting pair-maximality. □

Lemma 51 (EX: bounds at an edited hard pair)**.** *Let H be pair-maximal, S a solution with edited hard pair set U, $u = |U|$, let $xy \in U$ and let C be the cluster of x. Then:*

(a) if xy is deleted, $d_h(x) \leq D + \mathrm{del}_x + \mathrm{del}_y - 1$; if xy is inserted, $d_h(x) \leq D + \mathrm{ins}_y - 1 + \mathrm{del}_x$. In both cases $d_h(x) \leq D + u$;

(b) $|C| \leq 1 + D + d_h(x) + \mathrm{ins}_x$, hence $|C| \leq 1 + 2D + u$;

(c) every $z \in C$ has $d_h(z) \leq |C| - 1 + \mathrm{del}_z \leq 2D + 2u$.

Proof. (a) *Deleted:* $x \in C$, $y \notin C$. A hard neighbour z of x with xz not deleted ("kept") lies in C, and $z \neq y$. Apply Lemma 32 with centre x to the hard edges xz, xy: $zy \in E$ or zy is soft. If zy is a hard edge, it joins $z \in C$ to $y \notin C$, so it is a deleted hard edge at y other than xy (there are $\mathrm{del}_y - 1$ of these); otherwise $z \in Q(y)$, at most D vertices. So at most $D + \mathrm{del}_y - 1$ hard neighbours are kept, and $d_h(x) \leq D + \mathrm{del}_y - 1 + \mathrm{del}_x$. *Inserted:* $x, y \in C$. A kept hard neighbour z of x lies in C and $z \neq y$ (as $xy \notin E$); zy is a pair inside C, hence an edge of $G \triangle S$. If zy were a hard edge, $z \in N_h(x) \cap N_h(y)$, contradicting Lemma 50. So zy is soft ($z \in Q(y)$, at most D choices) or a hard non-edge, inserted at y and different from xy ($\mathrm{ins}_y - 1$ choices). So $d_h(x) \leq D + \mathrm{ins}_y - 1 + \mathrm{del}_x$. In the deleted case the pairs counted by $\mathrm{del}_x + \mathrm{del}_y - 1$ are distinct pairs of U, so the sum is $\leq u$; in the inserted case $\mathrm{ins}_y - 1 + \mathrm{del}_x \leq u - 1$ (an inserted pair at y other than xy and a deleted pair at x are distinct elements of $U \setminus \{xy\}$).

(b) For $z \in C \setminus \{x\}$ the pair xz is an edge of $G \triangle S$. Either xz is soft (at most D such z), or a kept hard edge (at most $d_h(x)$), or a hard non-edge, inserted at x (at most ins_x). With (a): in the deleted case $|C| \le 1 + D + (D + \mathrm{del}_x + \mathrm{del}_y - 1) + \mathrm{ins}_x \le 1 + 2D + u$ because the deleted pairs at x or y and the inserted pairs at x are distinct elements of U; in the inserted case $\mathrm{ins}_x + \mathrm{ins}_y - 1$ counts the inserted pairs at x or y and del_x is disjoint from them, so the total is again $\le u$.

(c) Hard neighbours of z inside C are at most $|C| - 1$; hard neighbours outside C are joined by deleted edges, at most $\mathrm{del}_z \le u$. Use (b). □

Slack 0 is attained in the finer forms of (a) and (b) in small cases (computer evidence), so the finer forms cannot be shaved by 1; the coarse bounds are loose.

Lemma 52 (F1: clusters of an optimum are connected)**.** *In an optimal solution every cluster C induces a connected subgraph $G[C]$.*

Proof. If $C = C_1 \cup C_2$ with $C_1, C_2 \neq \emptyset$ and no edge of G between them, splitting C into C_1, C_2 turns the $|C_1||C_2| > 0$ inserted pairs between them into non-edges outside S and creates no deletion. □

Fix ℓ and put

$$B = 2D + 3\ell, \qquad L_B = \{v : \ d_h(v) \le B\}, \qquad Z = B + D + 1 = 3D + 3\ell + 1.$$

In an optimal solution of size at most $|H| + \ell$ call a cluster *exceptional* if it contains an endpoint of a pair of U. Let BIG be the union of the hard components that contain a vertex v with $d_h(v) > B$.

Lemma 53 (NF: normal form)**.** *Let H be pair-maximal and S an optimal solution with $|S| \le |H| + \ell$, so $u \le \ell$.*

(α) There are at most $2u \le 2\ell$ exceptional clusters. Each exceptional cluster C has $|C| \le 1 + 2D + u \le 2D + \ell + 1$, lies in L_B, and $G[C]$ is connected.

(β) Let C be a non-exceptional cluster. Then no hard edge leaves C and no hard non-edge lies in C; so C is a union of whole hard components, each clean, and all pairs between different components inside C are soft. If C contains at least two components, $|C| \le 2D$.

(γ) Every hard component K of BIG *is a whole cluster of S, and this cluster is non-exceptional. Consequently every exceptional cluster is disjoint from* BIG*, and every component of* BIG *is clean.*

Proof. (α) Each pair of U has at most two clusters at its ends, which gives the count. The bound on $|C|$ is Lemma 51(b) (for an endpoint $x \in C$ of a pair in U), the inclusion $C \subseteq L_B$ is Lemma 51(c) together with $2D + 2u \le B$, and connectivity is Lemma 52.

(β) A hard edge leaving C would be deleted, with an endpoint in C; a hard non-edge inside C would be inserted. Both would make C exceptional. If K is a hard component meeting C, hard edges at a vertex of $K \cap C$ do not leave C, so $K \subseteq C$ by connectivity of K; K is clean since C contains no hard non-edge. Two vertices of different components inside C are not joined by a hard edge, and are not a hard non-edge, so their pair is soft. If C contains components $K_1 \neq K_2$, a vertex $x \in K_1$ has all vertices of $C \setminus K_1$ as soft partners, so $|C \setminus K_1| \le D$; likewise $|C \setminus K_2| \le D$; since $K_1 \subseteq C \setminus K_2$, $|C| = |K_1| + |C \setminus K_1| \le 2D$.

(γ) Let $d_h(v) > B$ and let $K = K(v)$ be its hard component. By Lemma 51(a) no endpoint of a pair of U has hard degree above $D + u \le B$, and by (α) no vertex of an exceptional cluster has hard degree above B. So the cluster C of v is non-exceptional, and by (β) $K \subseteq C$ and C is a union

of components. Now $|K| \geq d_h(v) + 1 > B + 1 > 2D$, so by (β) C cannot contain two components; hence $C = K$. Every other vertex of K lies in the same cluster. Cleanness is (β). The disjointness statement follows since a component of BIG is a non-exceptional cluster. □

Corollary 54 (CAND: bounded candidate partners)**.** *Let H be pair-maximal, S optimal with $|S| \leq |H| + \ell$, and $xy \in U$. Put $\rho = 2D + \ell$ and $\Delta = B + D$. Then $x, y \in L_B$ and:*

(i) if xy is deleted, $y \in N_h(x)$, and $|N_h(x)| \leq D + \ell$;

(ii) if xy is inserted, the distance from x to y in $G[L_B]$ is at most $|C| - 1 \leq \rho$, where C is the common cluster;

(iii) in both cases y lies in the ball $\mathrm{Ball}(x, \rho)$ of $G[L_B]$, which has at most $(\Delta + 1)^\rho = Z^\rho$ vertices.

The same holds with the roles of x and y exchanged.

Proof. $x, y \in L_B$ by Lemma 51(a) ($d_h \leq D+u \leq B$). (i) is the definition together with Lemma 51(a). (ii) $G[C]$ is connected (Lemma 52), $C \subseteq L_B$ by Lemma 53(α), so a shortest path in $G[C]$ has at most $|C| - 1 \leq 2D + u \leq \rho$ edges, and lies in $G[L_B]$. (iii) A vertex of $G[L_B]$ has at most B hard neighbours and at most D soft ones, so degree at most Δ, and a ball of radius ρ has at most $\sum_{i \leq \rho} \Delta^i \leq (\Delta + 1)^\rho$ vertices; deleted pairs have distance 1. Finally $\Delta + 1 = B + D + 1 = Z$. □

Remark 55 (Radius two). Lemma 39(b) gives more for inserted pairs: the ends of an inserted pair have a common neighbour inside the cluster, so their distance in $G[C] \subseteq G[L_B]$ is exactly 2, and (ii) holds with $\rho = 2$ in place of $2D + \ell$. This is an immediate combination of Lemma 39 and Lemma 53(α), and would replace ρ by 2 in Theorem 57 (and so $Z_*^{\rho_*\ell}$ by $Z_*^{2\ell}$); it was not part of the audited statements, so we keep the audited value in the theorems. (The computer checks observed distance ≤ 2 throughout.)

In the language of Section 8: Corollary 54 says that *one* endpoint of an edited hard pair determines a set of at most Z^ρ candidates for the other, for every pair, including (T2) pairs and deleted edges. What it does not bound is the number of candidates for the first endpoint, which is n.

9.3 Algorithms

Remark 56 (Fixed edited set). The proof of Theorem 6 shows: for a *fixed* set U' of hard pairs, $\min \sum_i |x_i|$ over all $x \in \prod \mathcal{D}_i$ with $G \triangle (U' \cup \bigcup x_i)$ a cluster graph is computed in $7^{w+1} n^{O(1)}$ time from a tree decomposition of $I(H)$ of width w, and for *every* U' of hard pairs the value $m_{U'} + |U'|$ is the size of an actual solution.

Theorem 57 (n^ℓ algorithm)**.** *Given (G, H, ℓ) and a tree decomposition of $I(H)$ of width w, CEaMP is solvable in time*

$$(n+1)^\ell \, Z_*^{\rho_*\ell} \, 7^{w+\ell+1} \, n^{O(1)}, \qquad D_* = D(H) + 2\ell, \quad Z_* = 3D_* + 3\ell + 1, \quad \rho_* = 2D_* + \ell.$$

Since $D(H) \leq 2(w+1)$, the factor $Z_*^{\rho_*\ell}$ is a function of (ℓ, w) only. This improves the exponent 2ℓ of Theorem 6 to ℓ; it is still XP in ℓ.

Proof. Compute M and $H' = H \cup M$ by Lemma 30; reject if $|M| > \ell$. Then H' is pair-maximal, $\ell' = \ell - |M| \leq \ell$, $D(H') \leq D + 2|M| \leq D_*$, a tree decomposition of $I(H')$ of width $w' \leq w + |M| \leq w + \ell$ is available, and the instance is equivalent to (G, H', ℓ'). All lemmas apply to H' with D_* (any upper bound on $|Q|$ is allowed) and ℓ'; put B, L_B, Z, ρ accordingly (they are at most B_*, Z_*, ρ_*).

Algorithm. Compute L_B and, for $x \in L_B$, the set $P(x)$ of vertices $y \in \mathrm{Ball}_{G[L_B]}(x, \rho) \setminus \{x\}$ with xy hard; $|P(x)| \le Z^\rho$. Enumerate all sequences of at most ℓ' pairs $(x_1, y_1), \ldots, (x_j, y_j)$ with $x_i \in L_B$ and $y_i \in P(x_i)$; there are at most $(1 + nZ^\rho)^{\ell'} \le (n+1)^{\ell'} Z^{\rho\ell'}$ of them. For each, let $U' = \{x_i y_i\}$ (a set of at most ℓ' hard pairs), compute $m_{U'}$ by Remark 56 for H' in $7^{w'+1} n^{O(1)}$ time, and accept iff $m_{U'} + |U'| \le |H'| + \ell'$.

Soundness. By Remark 56, $m_{U'} + |U'|$ is the size of an actual solution of G for every U' of hard pairs of H'; acceptance gives a solution of size $\le |H'| + \ell' = |H| + \ell$.

Completeness. Suppose the instance is a YES instance and let S be an optimal solution, $|S| \le |H'| + \ell'$, with edited hard pair set U, $|U| \le \ell'$ (Lemma 1 for H'). For each pair of U take one endpoint x; it lies in L_B and the other endpoint lies in $P(x)$ by Corollary 54. So U is among the enumerated sets, and for it $m_U + |U| \le |S| \le |H'| + \ell'$ (the packed part of S is a feasible choice of x for U by Lemma 1).

Time. $(n+1)^{\ell'} Z^{\rho\ell'} 7^{w'+1} n^{O(1)} \le (n+1)^{\ell} Z_*^{\rho_*\ell} 7^{w+\ell+1} n^{O(1)}$. □

Remark 58. The statement is in terms of the primed instance (G, H', ℓ'): $D_*, w+\ell$ and ℓ are upper bounds for $D(H'), w', \ell'$, which is the normalisation of Lemma 30. The exponent is halved because the candidate list is attached to *one* endpoint (Corollary 54); Theorem 6 guesses both. Note that ℓ bounds the *number* of edited hard pairs but not the cost of locating them: the n^ℓ comes from the position of the first endpoint of each pair. An FPT algorithm needs a bound on the positions as well. Computing the tree decomposition is FPT in w (Section 2).

Theorem 59 (FPT in (ℓ, D, ω))**.** *Let H be pair-maximal and let ℓ, D, B, Z, L_B,* BIG *be as above. Let $V' = V \setminus \mathrm{BIG}$ and let Γ' be the graph on V' whose edges are all edges of G and all soft pairs with both ends in V'; let $\omega = \mathrm{tw}(\Gamma')$ and assume a tree decomposition of Γ' of width ω is given. Then one can compute in time $(Z(\omega+1))^{O(\omega)} n + n^{O(1)}$ a number m^* such that*

$$\mathrm{OPT}(G) \le |H| + \ell \iff |E| + \sum_K f(K) + m^* \le |H| + \ell,$$

where K ranges over the hard components of BIG *and f is as in Lemma 49. Moreover $m_{\mathrm{all}} \le m^* \le m_{\mathrm{conn}}$, where m_{all} (m_{conn}) is the minimum of $\sum_{C \in \pi'} f(C)$ over all partitions π' of V' with clusters of at most Z vertices (and, for m_{conn}, connected in Γ').*

Proof. *($\Leftarrow$)* Let π' be the partition of V' realising m^* (the dynamic program below produces a real partition) and $\pi = \pi' \cup \{K : K \text{ a component of BIG}\}$. By Lemma 49(a), $\mathrm{cost}(\pi) = |E| + \sum_K f(K) + m^* \ge \mathrm{OPT}$, so the right-hand side gives $\mathrm{OPT} \le |H| + \ell$. No structure of G is needed for this direction.

($\Rightarrow$) Let S be optimal with $|S| \le |H| + \ell$. By Lemma 53(γ) every component of BIG is a whole cluster; let π' be the restriction of S's partition to V'. Its clusters are Γ'-connected and have at most Z vertices:

- *Exceptional C.* $C \cap \mathrm{BIG} = \emptyset$ by (γ), so $C \subseteq V'$; $G[C]$ is connected by (α) and its edges are edges of Γ'. $|C| \le 2D + \ell + 1 \le Z$.
- *Non-exceptional C not in* BIG. By (β), C is a union of whole hard components, none of which is in BIG (a component of BIG is a cluster by itself), so $C \subseteq V'$. Each component is connected by hard edges, which are edges of Γ'. If there are at least two components, all pairs between them are soft, hence edges of Γ', so C is Γ'-connected, and $|C| \le 2D$. If there is exactly one component $K = C$, it is clean, so every vertex y of K has all other vertices of K as hard neighbours or soft partners, $|K| \le d_h(y) + D + 1 \le B + D + 1 = Z$ because $y \notin$ BIG has $d_h(y) \le B$.

So π' is feasible for m_{conn} and $\mathrm{cost}(S) = |E| + \sum_K f(K) + \sum_{C\in\pi'} f(C) \geq |E| + \sum_K f(K) + m_{\mathrm{conn}} \geq |E| + \sum_K f(K) + m^*$.

The dynamic program. Take a nice tree decomposition of Γ' of width ω with empty root bag. A state at node t is a partition $\mathcal{P}$ of the bag X_t into blocks together with a size $s(\mathsf{b}) \in \{|\mathsf{b}|, \ldots, Z\}$ for each block b (the number of vertices introduced so far, in the subtree of t, of the cluster containing b). Its value is the minimum, over partitions of the introduced vertices into clusters that realise $(\mathcal{P}, s)$ (every cluster either contains a block of $\mathcal{P}$ or has been *closed*, i.e. has no vertex left in the bag), of $\sum_C g(C)$, with $g(C) = \binom{|C|}{2} - 2e(C)$ evaluated on the introduced vertices of C. *Introduce* v into the block b of size s (or into a new block of size 1): the value increases by $s - 2k$, where k is the number of G-neighbours of v in b; here we use that an introduced vertex y that has been forgotten cannot be a G-neighbour of v, because yv would be an edge of Γ' and y, v would share a bag. *Forget* v: if its block has other bag vertices the block shrinks; otherwise the cluster is closed and is never extended. *Join* of two children with the same partition $\mathcal{P}$: for a block with bag part b and sizes s_1, s_2 in the two children, put $x_1 = s_1 - |\mathsf{b}|$, $x_2 = s_2 - |\mathsf{b}|$; the new size is $s_1 + s_2 - |\mathsf{b}|$ and the value is $g_1 + g_2 - g(\mathsf{b}) + x_1x_2$ summed over blocks, where $g(\mathsf{b}) = \binom{|\mathsf{b}|}{2} - 2e(\mathsf{b})$ is read off the bag. This is correct because pairs inside b and edges inside b are counted in both children, pairs between the two exclusive parts number x_1x_2, and there is no G-edge between the exclusive parts (such an edge would be an edge of Γ' between vertices forgotten in different subtrees, which share no bag). *The join term x_1x_2 and the subtraction of $g(\mathsf{b})$ are essential*: omitting them gives a different value on some inputs (about 6% of the audit's tests). At the root the value is $\sum_C f(C)$ over a partition of V' with clusters of size $\leq Z$. Call it m^* (the minimum over states; we treat the closed clusters as final).

Every state sequence yields a real partition of V' with cluster sizes $\leq Z$ and correct cost, so $m^* \geq m_{\mathrm{all}}$. A Γ'-connected partition is represented: if the last bag vertex of a cluster C is forgotten at t, the vertex subtrees of C have a connected union (as C is Γ'-connected), which cannot contain the parent of t, so it lies below t and all of C has been introduced; hence no later vertex needs to join a closed cluster. So $m^* \leq m_{\mathrm{conn}}$. There are at most $(\omega+1)^{\omega+1}Z^{\omega+1}$ states per node, a join combines at most $Z^{2(\omega+1)}$ pairs of states with equal partition, and the tree has $O(\omega n)$ nodes; the total time is $(Z(\omega+1))^{O(\omega)}n$. Computing d_h, BIG and Γ' takes $n^{O(1)}$.

Combining: $\mathrm{OPT} \leq |H| + \ell$ implies $|E| + \sum_K f(K) + m^* \leq \mathrm{cost}(S) \leq |H| + \ell$ by ($\Rightarrow$), and the converse is ($\Leftarrow$). □

Remark 60 (Remarks on Theorem 59). (1) Cleanliness of the components of BIG need not be tested: it is implied when the instance is a YES instance (Lemma 53(γ)), and the inequality itself accounts for the true cost $f(K)$; rejecting an unclean component is a harmless optimisation. (2) The proof uses that an exceptional cluster is disjoint from BIG and hence lies in V'. It is *not* true that $L_B \subseteq V'$: a component of BIG can contain vertices of small hard degree (pendant vertices of a hub component); in the audit's tests this happened in 30 (instance, ℓ) pairs. (3) Here no pointer is guessed: far merges are carried by Γ'-connectivity. As for Theorem 57, D, B, Z, ℓ and ω refer to the pair-maximal instance (G, H', ℓ') of Lemma 30. (4) The dynamic program was implemented in the audit (introduce, forget and join over a min-fill decomposition, per connected component). With m computed by brute force, the criterion agreed with $\mathrm{OPT} \leq |H| + \ell$ in all 2,778 tested pairs of an instance and an ℓ ($n \leq 10$), and the DP value satisfied $m_{\mathrm{all}} \leq m^* \leq m_{\mathrm{conn}}$ in 2,141 checks; evidence, not proof. Without a tree decomposition of Γ', one is computed in $2^{O(\omega)}n$ time with width $2\omega + 1$ [10].

9.4 Why ω is not bounded by (ℓ, w)

Proposition 61 (ω is unbounded at $\ell = 0$, $w = 0$)**.** *For every cubic graph F there is an instance $(G_F, H_F, 0)$ with H_F a vertex-disjoint pair-maximal packing ($w = 0$, $c = 1$, $D = 2$), $\mathrm{OPT}(G_F) = |H_F|$ (a YES instance with $\ell = 0$), $\mathrm{BIG} = \emptyset$, every hard component of size at most 6 and every hard degree at most 4, and $\mathrm{tw}(\Gamma') \geq \mathrm{tw}(F)$. By [6] (cited, not re-proved here) there are cubic graphs F (expanders) with $\mathrm{tw}(F) = \Omega(|V(F)|)$. Hence ω is not bounded by any function of (ℓ, w).*

Proof. Orient each edge of F arbitrarily; for each oriented edge $e = (p \to q)$ introduce three vertices a_e, b_e, c_e. Let $K_v = \{a_e : e \text{ leaves } v\} \cup \{b_e, c_e : e \text{ enters } v\}$ and make each K_v a clique; add the cross edges $a_e b_e$ and nothing else. Then $|K_v| = \mathrm{out}(v) + 2\,\mathrm{in}(v) = 3 + \mathrm{in}(v) \leq 6$. $H_F = \{P_e = (a_e, b_e, c_e)\}$: $a_e b_e, b_e c_e \in E$ and $a_e c_e \notin E$ since $a_e \in K_p$, $c_e \in K_q$, $p \neq q$ and cross edges are of the form $a_{e'} b_{e'}$; so P_e is an induced P_3. The members are vertex-disjoint, so modification-disjoint, $c = 1$, $w = 0$ and $D = 2$.

Pair-maximality. In an induced P_3 $x\,y\,z$ not both edges can lie inside cliques, since then x, z would be in the clique of y (each vertex is in exactly one clique) and adjacent. So one of xy, yz is a cross edge $a_e b_e$, which is packed.

$\mathrm{OPT} = |H_F|$. The partition $\{K_v\}$ has no inserted pair and deletes exactly the cross edges, $|H_F|$ of them; and $\mathrm{OPT} \geq |H_F|$ by Lemma 1.

Hard structure. The soft pairs inside K_v are $b_e c_e$ (e entering v); all other pairs inside K_v are hard edges. So each K_v is one hard component and it is clean. The hard degree of $a_e \in K_p$ is $|K_p| - 1 \leq 4$ (as a_e is an out-edge, $\mathrm{in}(p) \leq 2$, $|K_p| \leq 5$); that of b_e or $c_e \in K_q$ is $|K_q| - 2 \leq 4$. So $d_h \leq 4 = 2D + 3\ell$ at $\ell = 0$, hence no vertex has $d_h > B$ and $\mathrm{BIG} = \emptyset$ (the margin is zero: one more unit would change the family).

Minor. $V' = V$, each K_v is connected in Γ' (a clique), and an edge or soft pair of Γ' between $K_p \neq K_q$ is one of $a_e b_e$, $a_e c_e$, which exist only for the oriented edge $e = (p \to q)$ of F. Contracting the K_v gives exactly F, so F is a minor of Γ' and $\mathrm{tw}(\Gamma') \geq \mathrm{tw}(F)$. □

Remark 62 (Scope of Proposition 61). The proposition refutes only the claim "$\omega \leq f(\ell, w)$"; it does *not* refute fixed-parameter tractability. The family has $w = 0$ and $\ell = 0$; it is solved in polynomial time by Theorem 6 and has $s = 3$ in Theorem 22. The obstruction is to the representation by Γ' (and by any single decomposition used both for the packed pairs and for the clusters): hard edges between far vertices are free (not in the budget), arbitrary in number, and make footprints of clusters disconnected in any decomposition of $I(H)$ or of W. At $\ell = 0$ they are harmless because the CSP of Theorem 5 needs no identities; for $\ell > 0$ an exceptional cluster can be assembled from vertices that are far apart (the example of a $c = 1$ instance with two vertex-disjoint P_3's joined by one hard edge that is deleted). So a dynamic program on one decomposition cannot be used for both purposes without a preprocessing step.

Remark 63 (Heuristic, not a theorem: colour coding does not repair the pointers). A greedy proper colouring of the ρ-th power of $G[L_B]$ with $(B + D)^\rho + 1$ colours names every vertex of a ball, so one site can verify a whole exceptional component by look-ups in G. What a site cannot do is make the other vertices of the same cluster, up to Z of them in far bags, declare the same cluster and agree on the variables of their P_3's: charging the cost at the minimum-colour vertex undercounts when that vertex takes the default state, and charging $1/|C|$ at each declaring site fails in the same way. Repairing this needs the identity of the other sites across bags, i.e. the n^ℓ guess. A sound variant exists for a decomposition of $I(H)$ plus all pairs of members within distance ρ in $G[L_B]$, which gives FPT in (ℓ, D, tw) of that graph, and by Proposition 61 its treewidth is not bounded in (ℓ, w). This is a heuristic analysis of a method, not a theorem.

9.5 The missing lemma

Conjecture 64 (ML: reduction lemma; missing)**.** *There is a polynomial-time algorithm that, given (G, H, ℓ) with H pair-maximal, either outputs NO correctly or outputs an equivalent instance (G^*, H^*, ℓ^*) with H^* pair-maximal, $\ell^* \le \ell$, $w(H^*) \le f_1(\ell, w)$ and $\mathrm{tw}(\Gamma'(G^*, H^*)) \le f_2(\ell, w)$.*

We do not know whether ML is true. An equivalent form needed for Theorem 59 is a rule that removes "inert" hard components (those intact in every solution and whose incident P_3's have a forced local solution). For $c = 1$ (a vertex-disjoint packing) the rule of [14] used in Section 6 plays this role and leaves $O(\ell)$ members; for $w \ge 1$ the counting breaks, because an edit outside all members can be an internal pair of another member, and the open case is that of Open Problem 5.

Observation 65 (Conditional; not a result)**.** *If Conjecture 64 holds, then Theorem 59 applied to (G^*, H^*, ℓ^*), with $D^* \le 2(w(H^*)+1)$, solves CEaMP in time $f(\ell, w)\, n^{O(1)}$, i.e. CEaMP is FPT in $\ell + w$.*

This is a one-line implication, stated only to record what ML would give; ML is a conjecture.

9.6 Computer evidence and status

The statements of this section were tested by brute force on small instances: in the notes, 5,300 instances with $n = 6, \ldots, 11$ and all partitions of cost at most $|H|+3$ ($> 4.5 \cdot 10^5$ partitions, $1.5 \cdot 10^6$ edited-pair checks, $>$ 17,000 optimal partitions) and an ILP on 100 instances with $n = 11, \ldots, 13$; in the audit, with independently written code, 1,511 instances with $n \le 10$ (8,988 optimal partitions). Result: Lemmas 49–53 and Corollary 54 had no violation, and inserted pairs had distance at most 2 in every test (consistent with Remark 55). The family of Proposition 61 was checked for K_4, $K_{3,3}$, the Petersen graph and random cubic graphs on 20 and 40 nodes (n up to 180): induced, pair-disjoint, pair-maximal, $\mathrm{cost}(\{K_v\}) = |H|$, hard degrees and component sizes as stated, and the quotient of Γ' by the K_v equal to F; for the four graphs F with three edges (not cubic; the construction does not use cubicity) on $n = 9$ vertices, in both orientations, $\mathrm{OPT} = |H|$ was verified over all partitions. $n \le 13$ is small; Proposition 61 is a construction, not an experiment. Evidence supports, and the proofs justify, the statements.

Statement	Status
Lemmas 49–53, Corollary 54	proved
Theorem 57 (n^ℓ, halved exponent)	proved, relies on Theorem 6, Lemma 30
Theorem 59 (FPT in (ℓ, D, ω))	proved (DP implemented only in the audit)
Proposition 61 (ω unbounded)	proved, expander treewidth cited [6]
FPT in $\ell + w$	**open**
Conjecture 64 (ML)	conjecture, not tested at size
Remark 63	heuristic

10 Limits of the parameterization

10.1 An ETH lower bound in the exponent

Theorem 66. *Assume ETH. There is no algorithm that decides CEaMP with $\ell = 0$ in time $2^{o(\mathrm{tw}(I(H)))} n^{O(1)}$. This holds even when every vertex lies in at most 23 packed P_3's (so $I(H)$ has maximum degree at most 66), and whether or not a tree decomposition is supplied.*

Proof. Li, Pilipczuk and Sorge [11, Thm. 1] give a polynomial reduction from 3-SAT to CEaMP with $\ell = 0$ and at most 23 packed P_3's per vertex. We use it as a black box together with its size accounting. After a linear preprocessing every clause has exactly three distinct variables and every variable occurs at least twice; let N be the number of variables and m the number of clauses, so $\sum_i m_i = 3m$ for the occurrence counts m_i. By the gadget descriptions of [11] (read in the audit, not recomputed constant by constant): a variable gadget has $4m_i$ cliques of 5 vertices and 25 packed P_3's per triple of consecutive cliques, hence $20m_i$ vertices and $50m_i$ packed P_3's; a clause gadget has $20 + 114$ vertices (clique sizes $1, 1, 4, 14$ and $34, 46, 34$), $6 + 12$ transferring P_3's, and $O(1)$ padding P_3's from a triangle packing on cliques of constant size. Hence $|V| \approx 194m$ and $|H| = O(m)$ with an absolute constant (not computed; a crude count gives about $3000m$). By the Sparsification Lemma (Impagliazzo, Paturi and Zane [8]; see also [4]) we may assume $m = O(N)$ at the cost of $2^{o(N)}$ extra time. Then $|H| = O(N)$ and $\mathrm{tw}(I(H)) \le |H| - 1 = O(N)$ (one bag). An algorithm with running time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$, which may compute its own decomposition, would therefore decide 3-SAT in time $2^{o(N)}$, contradicting ETH. □

Remark 67 (Caveats). (1) The bound is on the exponent only: it does not show that the base 3 (or 7) is tight, and $2^{o(w)}$ excludes $3^{o(w)}$ and $7^{o(w)}$ alike. A SETH-type bound $(3 - \varepsilon)^w$ is open. (2) The proof uses only $w \le |H| - 1$; it does not exhibit hardness for instances of small $|H|$ relative to w. (3) The constants of the size table were read from the reduction and not re-derived one by one; the structural claim (every gadget has $O(1)$ size per clause or occurrence) was re-derived. (4) The statement refers to the version-3 arXiv text of [11].

10.2 Other parameters do not suffice

Proposition 68. *CEaMP is NP-hard when $I(H)$ has no edge ($w = 0$, vertex-disjoint packing), with ℓ part of the input. Hence w alone is not enough, and (by Theorem 1 of [11]) neither is $\ell = 0$ alone.*

Proof. Cluster Editing is NP-hard (e.g. already on graphs of maximum degree 6 [9]). Given (G, k) compute greedily a maximal set H of vertex-disjoint induced P_3's (if none, G is a cluster graph). Vertex-disjoint implies modification-disjoint and $I(H)$ is edgeless. Since $\mathrm{OPT} \ge |H|$, (G, k) is a YES instance iff $(G, H, k - |H|)$ is (a NO instance if $k < |H|$). □

Proposition 69. $\mathrm{tw}(G)$ *and* $\mathrm{tw}(I(H))$ *are incomparable, in the following sense. (a) For the star $G = K_{1,2m}$ ($m \ge 1$) every packing $H \ne \emptyset$ has $I(H) = K_{|H|}$ and every YES instance (G, H, ℓ) satisfies $\ell + \mathrm{tw}(I(H)) \ge 2m - 2$, while $\mathrm{tw}(G) = 1$. (b) For the $k \times k$ grid G and a family H of vertex-disjoint induced P_3's along rows,* $\mathrm{tw}(I(H)) = 0$ *and* $\mathrm{tw}(G) = k$.

Proof. (a) Leaves have degree one, so the centre of any induced P_3 is the hub v; all packed P_3's are (a, v, b) with distinct leaves (the pairs va are disjoint), pairwise sharing v, so $I(H) = K_p$, $p = |H| \le m$, $\mathrm{tw} = p - 1$. If the hub's cluster contains j leaves, the cost is $(2m - j) + \binom{j}{2}$ (the other leaves are best left as singletons), which is minimized, with value $2m - 1$, at $j \in \{1, 2\}$; so $\mathrm{OPT} = 2m - 1$ and a YES instance has $\ell \ge 2m - 1 - p$, i.e. $\ell + \mathrm{tw}(I(H)) \ge 2m - 2$. (b) Rows of a grid are induced P_3's (no triangles) and $\mathrm{tw}(k \times k \text{ grid}) = k$ is standard. □

Thus hubs do not make $\mathrm{tw}(I(H))$ large by themselves; they force a tradeoff: small $\mathrm{tw}(I(H))$ on a star forces large ℓ.

11 Open problems

Notation as above; $w = \mathrm{tw}(I(H))$.

Open Problem 1 (Main question: (ℓ, w) in general)**.** Is CEaMP fixed-parameter tractable parameterized by $\ell + w$, i.e. solvable in $f(\ell, w)n^{O(1)}$ time, or is it W[1]-hard in $\ell + w$? Known: XP in ℓ and FPT in w for each fixed ℓ only via $n^{2\ell}$ (Theorem 6); FPT in ℓ when $w = 0$ by [14], and FPT in (ℓ, s) when the components of $I(H)$ span at most s vertices (Theorem 22); the case $H = \emptyset$ (plain Cluster Editing, $\ell = k$) is FPT. Note that adding c to the parameter changes nothing, since $c \leq w + 1$. Also known: $n^{\ell} f(\ell, w) 7^{w+\ell+1}$ time (Theorem 57) and FPT in (ℓ, D, ω) (Theorem 59), where ω is not bounded by (ℓ, w) (Proposition 61). A positive answer needs a bound $g(\ell, w)$ on the number of candidate *positions* of the first endpoint of each unpacked edit: given one endpoint, the other has at most $f(\ell, D)$ candidates (Corollary 54), so only the placement of at most ℓ bounded objects is free. Locality is proved for type T1 pairs, in particular soft-free merges (Corollary 41), but *not* for type T2 pairs, which include generic merges (Proposition 45). A negative answer has to encode $[n]$-valued information through type T2 pairs or deleted hard edges (a heuristic observation, Remark 48). What exactly remains is listed in Problems 2–4.

Open Problem 2 (Pointers of type T2)**.** Can the far ends of inserted pairs of type T2 (a kept hard edge az with a far from z in $I(H)$, and a soft pair zb; by Lemma 44 the end a is one of at most ℓ vertices fixed by the anchor (z, b)) be handled in a dynamic program of width bounded by $g(\ell, w)$? In particular, can a merge pointer between two pieces joined by a soft pair be forced at unbounded distance in $I(H)$? The computer evidence available (Remark 47) concerns only insertions inside one piece and is silent on this.

Open Problem 3 (Pointers realised by deleted hard edges)**.** The same question for deleted hard edges between far members of H (the other source of non-locality identified in Section 8.6). Given one endpoint, the other is one of at most $D + \ell$ hard neighbours (Corollary 54); the open point is the position of the first endpoint and the interaction of the far ends with the rest of the solution.

Open Problem 4 (The reduction lemma ML)**.** Is Conjecture 64 true: is there a polynomial-time reduction after which the treewidth ω of the graph Γ' of Theorem 59 is bounded by a function of (ℓ, w)? A positive answer would give FPT in $\ell + w$ (Observation 65). The family of Proposition 61 shows that a reduction is necessary; for $c = 1$ the rule of [14] is such a reduction, for $w \geq 1$ it is unknown.

Open Problem 5 ($w = 1$ with unbounded tree components)**.** If $I(H)$ is a forest ($w \leq 1$, hence $c \leq 2$) with components of unbounded size, is CEaMP FPT in ℓ? Theorem 22 settles bounded components, in particular perfect matchings (Corollary 23), but its counting argument is at the level of whole components and gives nothing for a single large tree component.

Open Problem 6 (Half-integral and intermediate packings)**.** Is CEaMP with $c = 2$ FPT in ℓ? What is the complexity of $\ell = 0$ for $3 \leq c \leq 22$? Both are open in [11], whose authors say a W[1]-hardness result for $c = 2$ "would also not be surprising".

Open Problem 7 (Base of the exponent)**.** For $\ell = 0$, does a $(3 - \varepsilon)^w n^{O(1)}$ (or even $2^{o(w)}$ with a smaller base) algorithm exist, or is there a SETH-type lower bound? Theorem 66 gives only the exponent.

Open Problem 8 (Treewidth of G)**.** What is the complexity of CEaMP with $\ell = 0$ (or of Cluster Editing) when $\mathrm{tw}(G)$ is bounded? We found no published result on Cluster Editing parameterized by treewidth alone (Section 12), so this must be proved rather than cited.

12 Related work

Packing lower bounds. van Bevern, Froese and Komusiewicz [14] parameterize F-free editing above the size of a vertex-disjoint packing; for Cluster Editing they obtain an $O(4^\ell \ell^3 + m + n)$-time algorithm for P_3 packings (Thm. 6.13), and for cost-t packings an $O(1.62^{(2t+1)\ell} + nm + n1.62^t)$ algorithm and a kernel with $O(t\ell)$ vertices. They show NP-hardness at $\ell = 0$ of K_3-free editing for edge-disjoint packings and of K_q-free editing, $q \geq 6$, for vertex-disjoint packings, and leave open Cluster Editing above edge-disjoint packings and the stronger modification-disjoint bound. Li, Pilipczuk and Sorge [11] answer the latter: NP-hard at $\ell = 0$ with at most 23 packed P_3's per vertex, $n^{2\ell+O(1)}$ for $c = 2$, and open for $c = 2$ in ℓ and for $3 \leq c \leq 22$ at $\ell = 0$. They report "grid-like structures" as the obstacle. Our $\mathrm{tw}(I(H))$ parameter does not appear in either paper.

Twins and normal forms. The twin lemma (Lemma 10) is the critical-clique property behind the kernelizations of Guo [7]; the closed-neighbourhood reduction rule that generalises it is Reduction Rule 6.2 of [14], which Section 6 applies to components of $I(H)$. Pair-maximality and the structure of R (Lemma 12) are our own bookkeeping; we make no priority claim.

Treewidth of expanders. Proposition 61 uses the fact that cubic expanders have treewidth linear in the number of vertices, which we cite from Grohe and Marx [6] (their relation between tree width, bramble size and expansion); we did not re-derive or re-check it.

Citations and treewidth. Of the citing works recorded by the notes, we opened the survey of Crespelle et al. [3], the vertex-splitting paper [13] (it uses modification-disjoint P_3 packings as a lower-bound tool), Bocci et al. [1] and Garvardt et al. [5]; none improves $c = 2$, settles $3 \leq c \leq 22$, or uses a treewidth-type parameter. Two further citing items (the PACE 2021 report and a 2021 thesis) were not opened, and citation coverage was incomplete (Semantic Scholar listed seven records, OpenAlex a count of two); this is a "not found", not proof of absence. We found no paper stating the complexity of plain Cluster Editing or Deletion parameterized by treewidth. The nearby s-Club Cluster Edge Deletion is FPT in $s + \mathrm{tw}$ but W[1]-hard in pathwidth [12]; as a deletion problem with a diameter bound it does not transfer to editing. Komusiewicz and Uhlmann [9] show that Cluster Editing is NP-hard at maximum degree 6, with no $2^{o(k)}$ algorithm under ETH even at bounded degree; this is consistent with the fact that bounded degree of $I(H)$ does not give tractability (Theorem 66).

References

[1] Bocci et al. A new temporal interpretation of cluster editing, 2022. Notes: opened. TODO: full author list.

[2] Hans L. Bodlaender. A linear-time algorithm for finding tree-decompositions of small treewidth. *SIAM Journal on Computing*, 25(6):1305–1317, 1996.

[3] Crespelle, Drange, Fomin, and Golovach. A survey of parameterized algorithms and the complexity of edge modification, 2020. Notes: opened.

[4] Cygan et al. *Parameterized Algorithms*. Springer, 2015. Notes (audit.md): recommended reference for dynamic programming on tree decompositions and for ETH lower bounds; the specific theorems were not opened.

[5] Garvardt, Morawietz, Nichterlein, and Weller. Graph clustering problems under the lens of parameterized local search. In *Proc. IPEC 2023*, volume 285 of *LIPIcs*, page Art. 20, 2023. Notes: opened (Dagstuhl PDF); the DOI is read off the recorded Dagstuhl URL. Journal version JCSS 2026 per the notes.

[6] Martin Grohe and Dániel Marx. On tree width, bramble size, and expansion. *Journal of Combinatorial Theory, Series B*, 99(1):218–228, 2009.

[7] Jiong Guo. A more effective linear kernelization for cluster editing. *Theoretical Computer Science*, 410(8–10):718–726, 2009.

[8] Russell Impagliazzo, Ramamohan Paturi, and Francis Zane. Which problems have strongly exponential complexity? *Journal of Computer and System Sciences*, 63(4):512–530, 2001.

[9] Komusiewicz and Uhlmann. Cluster editing with locally bounded modifications. *Discrete Applied Mathematics*, 2012. Notes: opened, `https://www.uni-marburg.de/de/fb12/arbeitsgruppen/algorith/paper/ce-bounded-degree.pdf`. Volume and pages not recorded in the notes.

[10] Tuukka Korhonen. A single-exponential time 2-approximation algorithm for treewidth. In *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 184–192, 2021. Crossref dates the proceedings 2022-02; the booktitle names the 2021 symposium.

[11] Li, Pilipczuk, and Sorge. Cluster editing parameterized above modification-disjoint P_3-packings. *ACM Transactions on Algorithms*, 20(1):Art. 3, 2023. Conference version: STACS 2021, LIPIcs 187, Art. 49. Notes: opened (arXiv v3). Surnames only, as recorded in the notes.

[12] Montecchiani et al. s-club cluster edge deletion, 2025. Notes: opened. Title and author list to be checked.

[13] Authors not recorded in the notes. The complexity of cluster vertex splitting and company, 2023. Notes: opened. TODO: fill in authors from the arXiv page.

[14] van Bevern, Froese, and Komusiewicz. Parameterizing edge modification problems above lower bounds. *Theory of Computing Systems*, 62(3), 2018. Conference version: CSR 2016. Notes: opened (arXiv).